



DETAILED PROJECT REPORT (DPR)

1 TPH MODULAR BIOCHAR PLANT

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6/8/26

DPR – Biochar Plant

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EXECUTIVE SUMMARY

1. EXECUTIVE SUMMARY

Project Overview

Shree Krishan CDR proposes the establishment of a 2×1 TPH Modular Biochar Plant at K R Pet Taluka, Mandya District, Karnataka – India’s first commercially scalable, digitally verified biochar-based Carbon Dioxide Removal (CDR) facility powered predominantly by renewable energy. The plant will process agricultural residues (coconut fronds, paddy straw, and sugarcane trash) sourced within a 40 km radius, converting them into stable, high-carbon biochar through a continuous, oxygen-free pyrolysis process.

Key Project Parameters

Promoter: Shree Krishan CDR | Location: K R Pet Taluka, Mandya District, Karnataka, India | Plant Capacity: 2×1 TPH continuous pyrolysis lines = 4,800 TPA biochar | Feedstock Requirement: ~14,000 TPA (as-received basis) of agricultural residues | Total CAPEX (Module 1): Rs 4.2 Crore (approx. USD 500,000), excluding land | Target Commissioning: Q4 2026 (October–December 2026) | CDR Potential: 9,360–9,750 tCO₂ per module per year | Revenue Model: Direct B2B offtake agreements for certified carbon removal credits; supplementary revenue from bio-oil, wood vinegar, and biochar sales.

Strategic Rationale

K R Pet Taluka presents an ideal combination of feedstock abundance, agrarian community engagement, and logistics efficiency. The taluka’s 91,525 Ha of agricultural land generates substantial volumes of coconut fronds, paddy straw, and sugarcane trash – currently disposed of by open burning. The project transforms this waste into a permanent carbon sink while returning agronomic value to farmers through a barter model: biochar and wood vinegar in exchange for biomass feedstock, creating a self-sustaining circular carbon economy.

Scale-up Vision

Our modular biochar plant is designed for “Efficient utilization of agri-waste, our immediate focus is to optimize biomass by effective drying leading to more of carbon removal and carbon offset for carbon trading. Module 1 (Q4 2026) is the pilot and MRV baseline unit. Six modules are planned over 36 months, reaching a combined capacity of 28,800 TPA biochar and removing over 40,000 tCO₂ annually by FY 2029-30. The Standardized Modular Management Model (SMMM) ensures that each successive unit is deployed faster and at lower cost, with identical components, shared spare parts inventory, and centralized digital MRV oversight. The entire value chain – from village-level biomass collection to carbon credit issuance – is digitally tracked via a blockchain-enabled MRV platform, ensuring high integrity and auditability for B2B corporate buyers.

CDR PROJECTS OVERVIEW

2. CDR PROJECTS OVERVIEW

A. Biochar Sequestration

Feedstock Yield and Sequestration Metrics

To arrive at the ton-based metrics for coconut fronds, the following specific data points were used:

- **Coconut Yield Formula:** An adult palm produces **12 to 14 leaves annually**. At a standard tree density of **175 trees/Ha**, this equals ~2,100 to 2,450 fronds. Using the air-dry weight of **~2.65 kg per whole leaf**, the yield is calculated as **5.56 to 6.49 MT per hectare**.
- **Paddy Straw Yield Formula:** **4 tons per acre**, converted to hectares (1 Ha = 2.47 Acres), the yield is **9.88 MT per hectare**.
- **Bamboo Productivity:** High-yield **Bheema Bamboo** is utilized to create a lasting carbon sink, with modular plots developed for both production and sequestration.

Operational Supply Strategy (4,800 TPA Biochar Facility)

To ensure raw material stability, the plant uses a **Mixed Feedstock Strategy** to mitigate seasonal "Nil" periods:

- **Safety Buffer: Coconut fronds** act as the primary year-round baseline feedstock (40% of mix) because they are available **365 days a year**.
- **Seasonal Peaks: Sugarcane trash** (August–March) and **Paddy straw** (March–May and Sept–Nov) are prioritized during harvest windows to maximize field-level sequestration.
- **Sequestration Targets:** The project aims to cover **3,000 hectares** over 3 years to sequester **12,000 tons of biochar** at 80% efficacy.

Monthly Supply & Operational Strategy

To maintain a steady **5,000 TPA biochar production**, the plant follows a seasonal feedstock strategy. This chart reflects the monthly availability of primary field residues versus baseline year-round feedstocks.

Month	Sugarcane Trash	Paddy Straw	Coconut Fronds	Operational Focus
December	●	●	●	Peak harvest: Processing cane and paddy residues.
January	●	●	●	Peak harvest: Maximum field residue collection.
February	●	●	●	Peak harvest: Final winter residue aggregation.

March	●	●	●	Peak Supply: Dual-crop availability (Cane + Paddy).
April	●	●	●	Stockpiling: Shifting to stored buffer residues.
May	●	●	●	Transition: Processing summer paddy and stored cane trash.
June	●	●	●	Monsoon Gap: High reliance on Coconut Fronds .
July	●	●	●	Monsoon Gap: Baseline operations using pulverised fronds.
August	●	●	●	Buffer Use: Processing stored materials and fronds.
September	●	●	●	Kharif Start: Early paddy straw and cane tops.
October	●	●	●	Kharif Peak: Resuming full-scale field residue processing.
November	●	●	●	Kharif Peak: High-volume sequestration in harvested fields.

Legend: ● = Peak Availability; ● = Moderate/Buffer Supply; ● = Low/Nil (Reliance on Storage)

Table 1: Modular Plant Annual Mass Balance & Carbon Potential
(Based on 13,000 TPA Dry Feedstock)

Category	Item	Value / Volume	Key Remark
Input	Dry Chipped Biomass	13,000 TPA	Targeted feedstock source < 50 km
Product	Biochar (35% yield)	4550 TPA	72–75% Carbon content
Bi-product	Bio-Oil (25% yield)	3,250 TPA	Used as repellent & Bio-binder
Bi-product	Pyrogas (40% yield)	5,200 TPA	Partially used for heating or other process
Sequestration	Biochar Component	0.60 – 0.64 tCO ₂	Per ton of dry feedstock
Sequestration	Bio-Oil Component	0.12 – 0.15 tCO ₂	Permanent storage in asphalt
Total Impact	Annual CO₂ Removal	9,360 – 9,750 tCO₂	Gross capacity per module

B. NBS for CO2 removal

Bheema Bamboo Strategy

The NBS strategy utilizes the rapid growth and high carbon-sequestration capacity of **Bheema bamboo** to create long-term "carbon sinks". This project follows a phased implementation model integrated with regenerative agriculture to ensure early cash flows and maximum ecological impact.



1. Phased Development Model

The project scales from an initial start-up phase to a significant district-level presence by 2030.

- **Start-up Phase:** Development of **150 hectares** across five Talukas (including KR Pet), distributed over approximately 400 plots. Each Taluka is targeted for **3,000 TPA** of carbon removal capacity.
- **By 2030:** Expansion to **1,000 hectares** (approximately 2,000 plots) with a capacity of **20,000 TPA per Taluka**. At this stage, bamboo-derived biochar will also be supplied to industrial sectors like steel plants.

2. Integrated Horticulture & Inter-cropping

To maintain financial viability during the initial growth years (Years 1–3), the strategy adopts a modular **Horticulture Project model** in 3-hectare parcels:

- **Dense Bamboo Cultivation:** Approximately **20,000 sq. m** dedicated to Bheema bamboo, expected to yield **200 TPA from the 4th year**.
- **High-Value Inter-cropping:** During years 1 and 2, medicinal and high-value vegetables (e.g., ginger) are cultivated between bamboo rows.
- **Protected Cultivation:** Integration of **2,000 sq. m Polyhouses** for salad greens and cruciferous vegetables, yielding **60 TPA from the first year**.

3. Carbon Removal & "Permanence" Standards

Bheema bamboo groves can live for up to **100 years**, allowing for annual selective harvesting without damaging the total carbon stock. To generate premium carbon credits, the project adheres to strict protocols:

- **Validation & Verification:** Alignment with international standards such as **Verra (VCS) / Gold Standard (GS)**
- **Permanent Storage:** Credits are issued based on the harvested bamboo being used in **long-lived applications** to prevent immediate CO2 release.
 - **Durable Products:** Construction materials (beams, poles) and engineered products (flooring, plywood).
 - **Biochar:** Conversion into biochar for soil incorporation, ensuring carbon remains sequestered for centuries.
- **Digital MRV:** Utilization of a high-integrity, **blockchain-enabled MRV framework** for village-level origin tagging and continuous mass balance tracking.

4. Social and Ecological Outcomes

The strategy is managed by skilled **green entrepreneurial gangs** mentored by **GAM (Grameena Abhivrudhi Mandall)**, a Section 8 company.

- **Annual Removal Targets (2026-27):** Targeted removal of **15,000 tCO2e** across KR Pet and Mandya district.
- **By 2030-31:** Aiming for a total of **60,000 tCO2e** carbon removal annually.
- **Biodiversity Enhancement:** Agro-forestry initiatives on bunds include secondary species like Lime, Moringa, and Pongamia to enhance the local ecosystem.

BIOCHAR OVERVIEW

3. BIOCHAR OVERVIEW

A. Product

Biochar, a carbon-rich byproduct of biomass pyrolysis, enhances soil health by improving fertility, structure, water retention, and microbial activity while enabling carbon sequestration. It also acts as an adsorbent in non-soil applications, including water/air filtration, environmental remediation, and construction materials like concrete, offering a sustainable, versatile solution for carbon storage.

Soil Applications of Biochar

- a. **Improved Soil Fertility:** Biochar boosts nutrient availability and nitrogen retention, which can increase crop yields, particularly in poor, degraded soils.
- b. **Water Management:** High porosity increases water holding capacity, enhancing water retention and reducing nutrient leaching.
- c. **Soil Structure & Health:** It enhances soil aeration, reduces soil compaction, and stimulates beneficial microbial activity.
- d. **Environmental Remediation:** Biochar adsorbs heavy metals and organic contaminants, helping to clean up polluted, saline-alkali, or industrial soils.
- e. **Carbon Sequestration:** As a stable, high-carbon material, it locks carbon in the soil for decades, allowing for carbon credits.

Non-Soil Applications of Biochar

- a. **Environmental Filtration:** Used for water and air purification, absorbing contaminants and odors due to its large surface area.
- b. **Construction Materials:** Incorporated into asphalt, concrete, and plasterboard to improve properties and create carbon-negative products.
- c. **Energy Storage:** Utilized in the production of supercapacitor electrodes for storing electric charge.
- d. **Industrial Absorbents:** Used in various industrial processes as a cost-effective, sustainable alternative to activated carbon.

B. Process

1. Carbon Stability & Permanence

Biochar contains large fractions of aromatic carbon with slow oxidation rates. Stability is influenced by Pyrolysis temperature (500–550°C), Feedstock lignin content (high in coconut and bamboo) and Reactor residence time

The project's design ensures Stable carbon fraction of 60–80%, Permanence >100 years; median 300–500 years and in line with the methodology requirements of Verra VCS

2. Advanced Pyrolysis Technology

Technology Characteristics with Oxygen-less continuous pyrolysis, Controlled residence time and temperature with High thermal efficiency and Stable carbon output consistency

3. Biochar Characteristics

- Highly porous and nutrient-rich: Enhances soil structure, water retention, and nutrient availability.
- High carbon content: Stabilizes and sequesters carbon, aiding in long-term carbon storage.
- Alkaline pH and pollutant adsorption: Neutralizes acidic soils and binds contaminants, improving soil health and remediation.

4. Integrated Pre-Processing and Quality Control Systems

Pre-Processing Infrastructure includes Shredding, chipping, and pulverizing systems, drying zones for moisture control and grinding for uniform particle size preparation

Quality Assurance includes checking biomass / agro-residues feedstock moisture content at source, Batch-wise sampling at plant, Laboratory testing (C%, moisture, ash) & Rejection and re-processing protocols

*These systems ensure **repeatable, verifiable, and auditable carbon outcomes**, a key differentiator for premium carbon credits.*

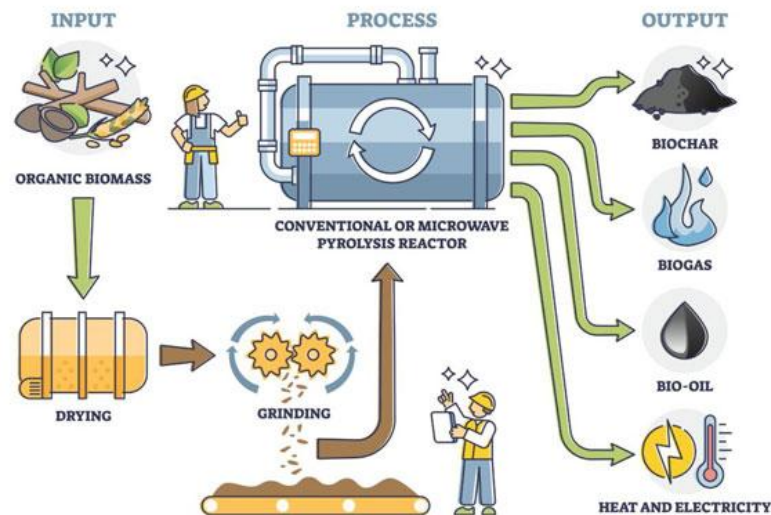
5. Robust MRV and Digital Traceability Framework

End-to-End Traceability ensures Village-level biomass origin tagging, QR-coded batch identification and provision for GPS-tracked transport logs

MRV Compliance for Continuous mass balance tracking, Carbon fraction verification & Emissions monitoring and reporting

Blockchain Integration for Immutable data records, Transparent audit trails and Enhanced buyer confidence

This digital MRV framework positions the biochar plant as a best-in-class carbon removal project.



A. Applications

The biochar produced is designed for high-integrity applications across diverse sectors, leveraging its stable carbon properties and high surface area.

A. Agricultural & Horticultural Applications

- **Soil Amendment:** Biochar restores optimal soil organic carbon (SOC) levels, particularly in the nutrient-depleted soils.
- **Water Retention:** Enhances the soil's water-holding capacity, providing resilience against seasonal dry spells for crops like sugarcane and paddy.
- **Nutrient Delivery:** Functions as a carrier for fertilizers, improving nutrient-use efficiency and reducing chemical runoff.
- **Regenerative Farming:** Integrated into "environment farming" for high-nutrition and medicinal vegetables within the project's Controlled Environment Agriculture (CEA) framework.

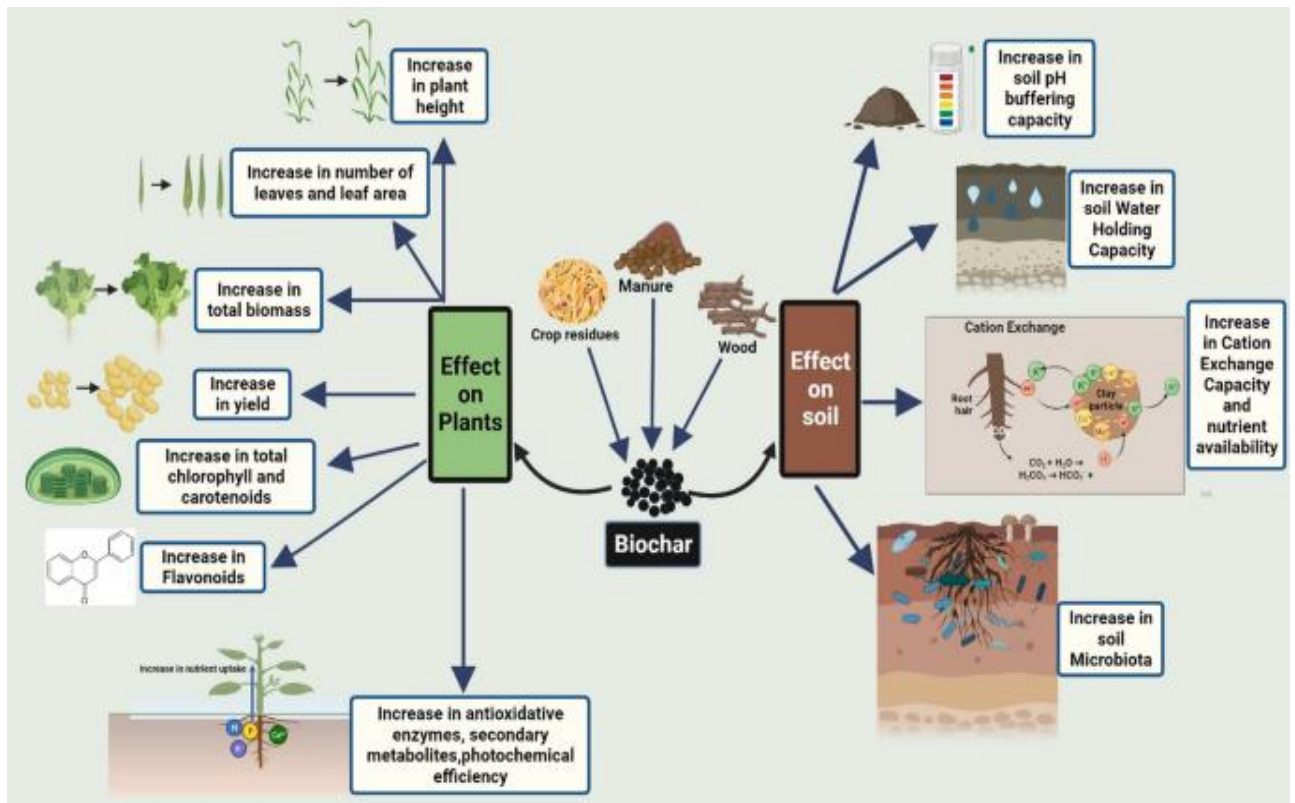


Image courtesy: <https://link.springer.com/article/10.1007/s44279-024-00033-2/figures/1>

B. Industrial Decarbonization

- **Green Steel Production:** Biochar serves as a sustainable replacement for imported coking coal and met-coal in the steel industry.
 - **Iron Ore Pellets:** Used as a reductant and fuel in the pelletization process.
 - **Electric Arc Furnaces (EAF):** Injected into furnaces to reduce the carbon footprint of steel manufacturing.
- **Binder Applications:** By-products such as **Bio Binder** (a thick residue post-decantation) find application as renewable binders for road construction and industrial briquettes.

C. Environmental Remediation

- **Contaminant Filtration:** Due to its high surface area and porous structure, biochar is effective in water filtration and heavy metal removal.
- **Soil Remediation:** Used to treat contaminated industrial or agricultural land by immobilizing pollutants and restoring biological health.

D. Carbon Removal Credits (CDR)

- **High-Integrity Sequestration:** The primary "application" for the investor market is the generation of certified carbon removal credits.
- **Permanence:** Provides a verifiable carbon sink with a validated storage duration of **100 to 1,000 years**, meeting strict international standards such as Verra, GS, and Puro.

Summary Table

Market Segment	Primary Application	Key Value Proposition
Agriculture	Soil Organic Carbon (SOC) enhancement for moisture retention	6% increase in sugarcane yields; water resilience
Steel Industry	Coking coal replacement	Alignment with Green Steel Taxonomy; decarbonization
Infrastructure	Bio-binder for road construction	Use of by-products for circular economy
Carbon Market	CDR Credit generation	High permanence (>100 years); blockchain MRV

PROJECT OBJECTIVES & MODULAR DESIGN

4. PROJECT OBJECTIVES & MODULAR DESIGN

A. Human Capital Creation (Capacity building)

The project's sustainability is predicated on the development of a skilled local workforce and the incubation of rural green enterprises. By fostering human capital, the project ensures that the transition to a circular carbon economy is inclusive and community-led.

In collaboration with the Skill Council for Green Jobs (SCGJ) and its Centre of Excellence (CoE), the project will establish dedicated Skilling and Green Economy Startup Incubation Centre in K.R. Pet Taluka, Mandya District. This centre will serve as the nerve centre for:

- **Green Entrepreneurship:** Identifying and mentoring local youth to manage the biomass value chain.
- **Supply Chain Management:** Specialized training in feedstock collection, mechanical aggregation, pre-processing (chipping/powdering), and the logistics of biochar sequestration.
- **Operational Excellence:** Technical training for plant operators to manage the Biochar pyrolysis units and digital MRV systems.

B. Supply Chain Management

The project employs a decentralized "Hub-and-Spoke" model for biomass procurement, focusing on rural entrepreneurship and rigorous quality control to ensure a steady supply of high-quality feedstock.

C. Carbon Cycle Management

The project adopts a multi-pronged approach to Carbon Dioxide Removal (CDR) and emissions abatement, integrating Nature-Based Solutions (NBS) with advanced thermochemical conversion and engineered storage.

a. Carbon Dioxide Removal (CDR): Biochar & Bio-Oil Sequestration

The primary carbon sink is established through the pyrolysis of agricultural residues.

- **Biochar Pathway:** By converting biomass into biochar, carbon that would have otherwise returned to the atmosphere through decomposition or burning is stabilized for centuries.
- **Bio-Oil & Bio-Binder:** The condensed pyrogas fraction (Bio-oil) is processed into a high-density bio-binder. Its integration into asphalt and road infrastructure constitutes a permanent geological-style storage pathway, preventing the re-release of carbon through oxidative degradation.

b. Nature-Based Solutions (NBS): Bamboo Carbon Sinks

The project integrates managed **Bheema Bamboo plantations** as a high-velocity biological carbon pump. Bamboo exhibits one of the highest sequestration rates among terrestrial plants. By utilizing biochar to enhance the Soil Organic Carbon (SOC) of these plantations. Bamboo as biomass is planned as a long-term solution as bamboo needs minimum of 3-4 years for full growth & to be used as biomass.

c. Integrated Digital MRV & Sustainable Farming

To ensure the integrity of the carbon cycle, every node of the supply chain is digitized.

- **Transparency:** The digital Monitoring, Reporting, and Verification (MRV) platform tracks the biomass from collection through to soil application.
- **Sustainable Agriculture:** This system is integrated directly with farm-level management, ensuring that carbon sequestration also drives "Sustainable Farming" outcomes, such as reduced chemical fertilizer dependence and improved water retention.

D. Business model:

The project operates on a Standardized Modular Management Model, focusing on rigorous process control, asset-heavy centralized oversight, and symbiotic relationships with landholders. This framework ensures rapid scalability while maintaining the high-quality carbon removal (CDR) standards required for international certification.

a. The Standardized Operational Framework

The expansion strategy is built upon a systems-based approach that emphasizes absolute uniformity in output through decentralized operations governed by a central authority.

- **Asset & Site Control:** To ensure long-term stability and operational integrity, the core entity secures land via long-term leases for the placement of pre-processing units, structural sheds, and auxiliary equipment.
- **Integrated Infrastructure:** To eliminate variability in feedstock quality, the project provides and maintains all critical machinery—including high-capacity shredders, pulverisers and chippers. This ensures that the pre-processing process remains consistent across all decentralized nodes.
- **SOP Adherence & Quality Assurance:** A comprehensive set of Standard Operating Procedures (SOPs) governs every phase of production. Strict adherence to these protocols is mandatory, ensuring that the biochar and bio-oil produced are chemically and physically identical and satisfying the "comparability" requirements of digital MRV verification.

b. Precision Harvesting and In-Situ Logistics

To minimize the carbon footprint and maximize the efficiency of feedstock collection, the model employs a specialized logistics protocol tailored for plantation environments:

- **Navigational Efficiency:** Collection operations utilize **tiller trolleys** specifically selected for their narrow wheelbase and high manoeuvrability. This allows for seamless navigation through the specific spacing gaps found in commercial coconut, paddy & cane plantations without damaging the standing crop.
- **Standardized Material Handling:** SOPs govern the systematic raking and loading of biomass into moisture-proof **tarpaulin systems**. This prevents soil contamination and ensures the biomass remains at optimal moisture levels for immediate pre-processing.

c. The Barter Value Proposition (Circular Economy)

The business model establishes a "win-win" barter transaction framework with plantation owners, which reduces upfront cash requirements and fosters deep community integration:

- **Service Exchange:** The project provides professional biomass clearing services, removing dry coconut fronds and agricultural residue directly from the field. This offers immediate economic value to the owner by eliminating the labor costs associated with waste management and significantly reducing fire hazards within the plantation.
- **Product Incentives:** In direct exchange for the biomass feedstock, the plantation owner receives Wood Vinegar (a potent organic bio-stimulant) at no cost. This is supplied in standardized 25 kg reusable drums, promoting sustainable packaging practices.
- **Recurring Revenue & Farmer Support:** To encourage the long-term adoption of regenerative practices, additional volumes of wood vinegar are supplied to participating farmers at a highly concessional rate of Rs 80/kg. This creates a recurring demand loop for the project's liquid outputs while enhancing the farmer's productivity.

Manual Raking



Tarp Collection



Pre-processing

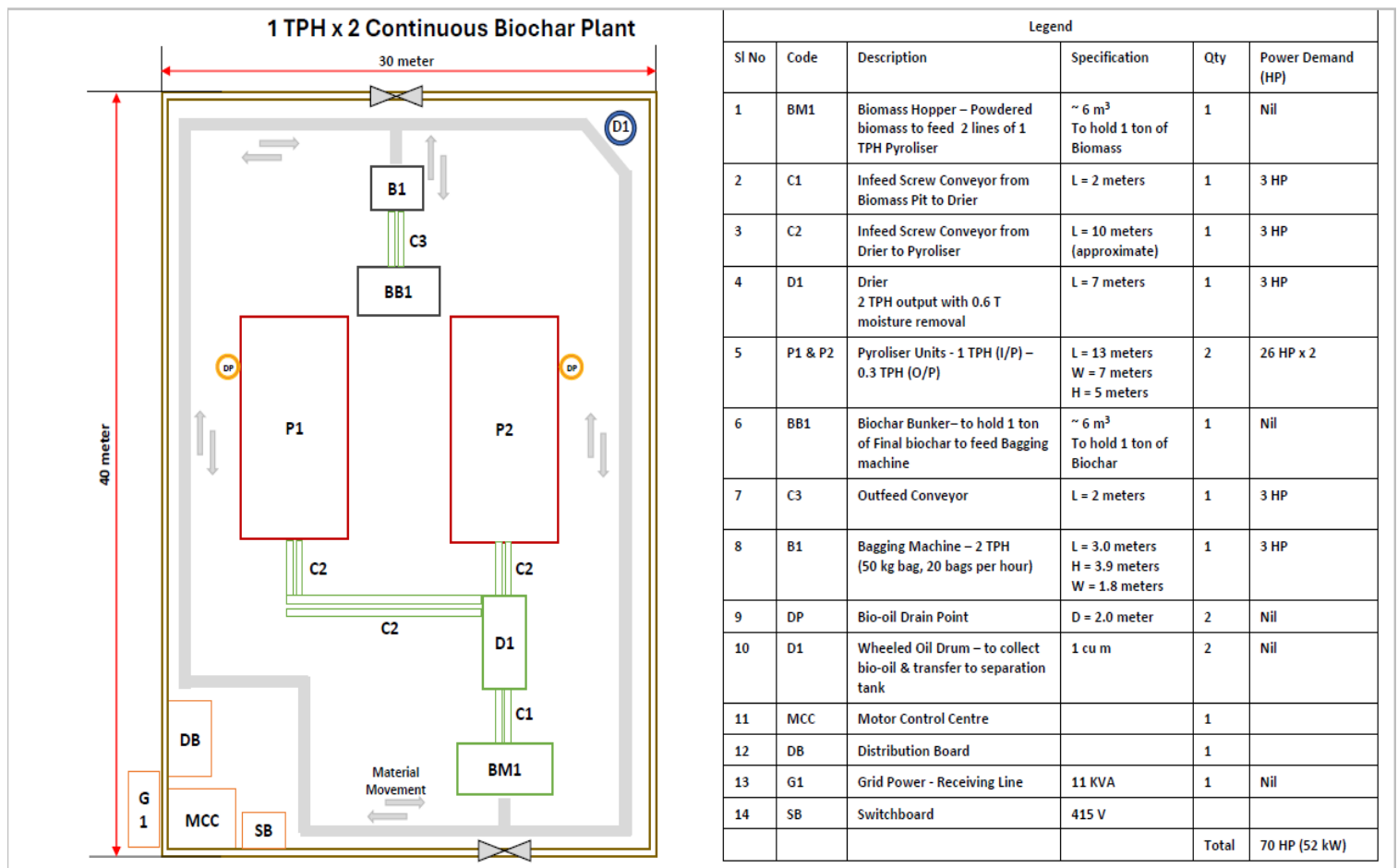


Transportation – Source to Plant biomass storage





E. Modular Design:



***PROJECT LOCATION;
SITE MAP & PLOT
PLAN***

5. PROJECT LOCATION; SITE MAP & PLOT PLAN

The proposed Modular Biochar Project is located in **K R Pet taluka**, Mandya region, Karnataka, India. The project site has been strategically selected based on feedstock availability, logistical efficiency, regulatory compatibility, environmental suitability, and socio-economic considerations.

K R Pet taluka forms part of the southern agro-ecological belt of Karnataka and is characterized by mixed cropping systems, plantation agriculture, and proximity to agro-processing infrastructure such as sugar mills, jaggery units, and coconut-based value chains. These characteristics make the location particularly suitable for a large-scale biochar project dependent on diverse agricultural residues.

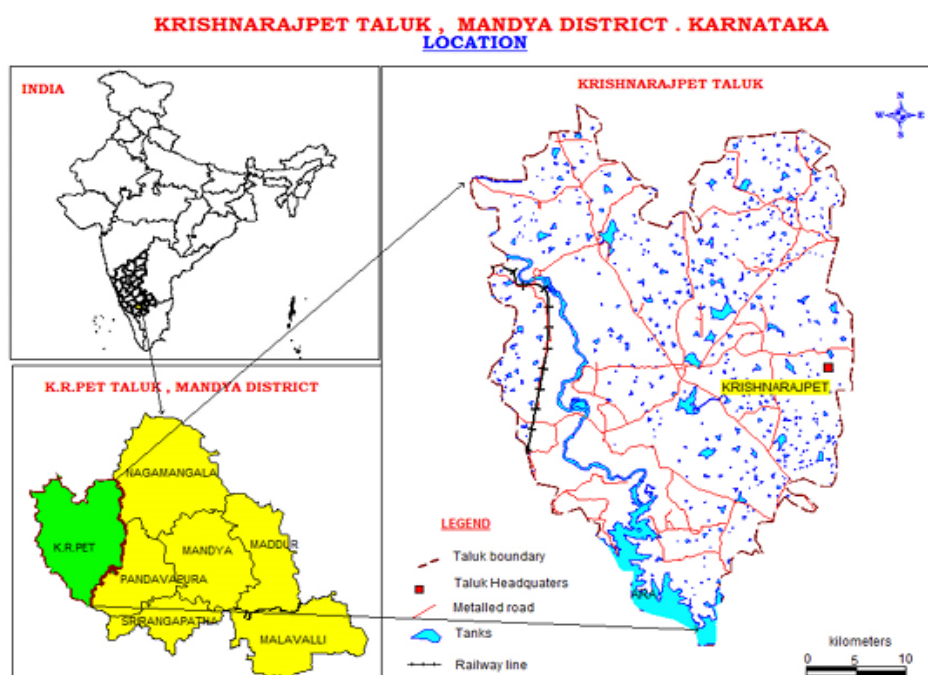
The site supports the project's dual objectives of:

- Establishing a technically viable and financially sustainable biochar facility, and
- Creating a decentralized rural biomass aggregation ecosystem for long-term carbon sequestration.

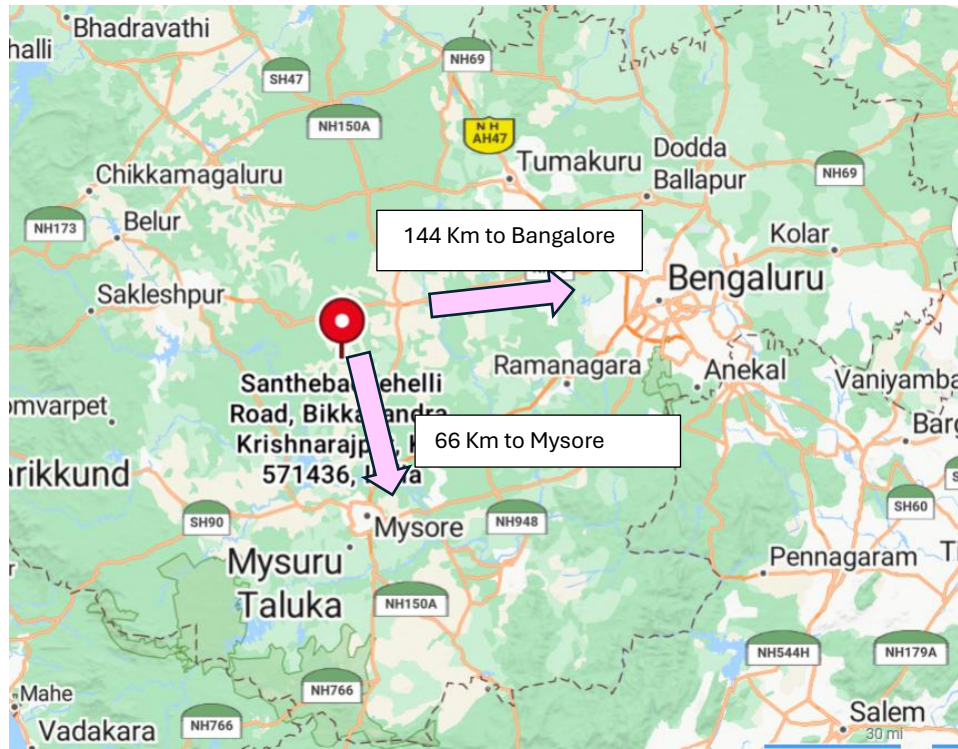
A. Geographic and Administrative Details

- **Country:** India
- **State:** Karnataka
- **Region:** Mandya (administrative region)
- **Taluka:** K R Pet
- **Nearest Urban Center:** K R Pet town (administrative and logistics hub)
- **Nearest District Headquarters:** Mandya
- **Nearest Major City:** Mysuru / Bengaluru (regional markets and logistics access)

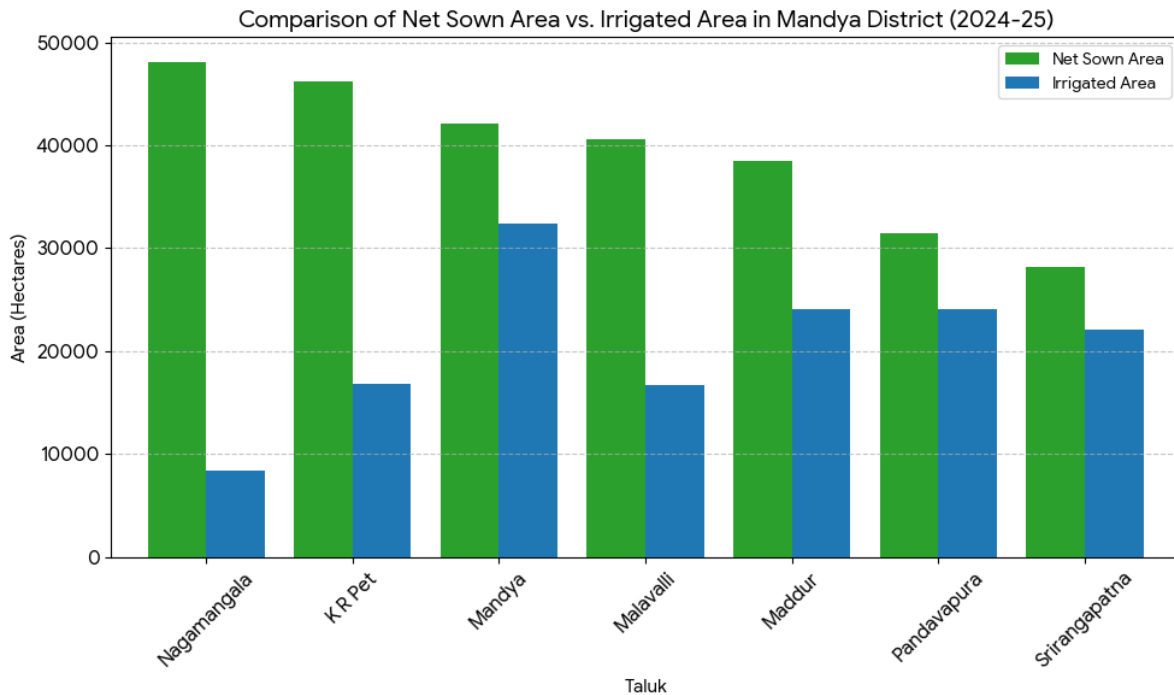
The project location lies within an established rural governance framework, enabling coordination with local authorities for land use, factory licensing, pollution control, and labor compliance.



Map Source: <https://antharjala.karnataka.gov.in/storage/pdf-files/NAQUIM%20REPORTS/120.pdf>



Land Use Statistics: Mandya District (FY 2024-25)*							
Taluk	Total Geographical Area (Ha)	Net Sown Area (Ha)	Irrigated Area (Ha)	Forest Area (Ha)	Pasture / Wasteland (Ha)	Cultivable Area (%)	Notes
K R Pet	91,525	46,244	16,812	4,210	12,410	50.50%	High pulse cultivation
Pandavapura	53,842	31,422	24,115	2,145	6,340	58.30%	Major Sugarcane belt
Nagamangala	1,03,450	48,120	8,420	6,850	25,140	46.50%	Higher dryland/pasture
Mandya	70,500	42,150	32,450	1,510	8,920	59.80%	Most fertile/irrigated
Srirangapatna	43,212	28,140	22,110	1,825	4,120	65.10%	Max canal irrigation
Maddur	64,215	38,450	24,120	1,940	9,815	59.80%	High Paddy intensity
Malavalli	71,500	40,541	16,715	6,285	15,662	56.70%	Critical groundwater
Total (Dist)	4,98,244	2,75,067	1,44,742	24,765	82,407	55.20%	<i>District Totals</i>



* Based on the latest available official data from the **Mandya District Administration** and the **Karnataka State Department of Agriculture** (as of the 2024-25 reporting cycle), here is the populated table for the seven taluks of Mandya district.

B. Site Coordinates and Physical Characteristics

The selected site lies within the central zone of K R Pet taluka.

Indicative coordinates (to be finalized):

- **Latitude:** ~12.51° N (12.789235781699842)
- **Longitude:** ~76.52° E (76.57098373323251)

C. Site Area and Land Characteristics

- **Total land area:** Approximately 4.0 acres (owned), ~5 acres (long-term leased)
- **Topography:** Flat to gently sloping terrain
- **Elevation:** ~700–750 meters above mean sea level
- **Soil Type:** Red loamy to lateritic soils with moderate bearing capacity
- **Flood Risk:** Low; site located outside flood-prone zones
- **Seismic Zone:** Zone II (low seismic risk)

The land is suitable for industrial development with minimal earthwork and no requirement for major land reclamation.

D. Feedstock Catchment and Spatial Distribution

The project's location provides access to a **multi-village biomass catchment zone** covering approximately **90 villages within K R Pet taluka**.

Primary Agricultural Residues

- Coconut fronds and shells
- Sugarcane trash and bagasse
- Mixed dry agricultural residues

Catchment Radius

- Primary sourcing radius: 0–25 km
- Secondary buffer radius: 25–40 km (used during seasonal peaks)

The decentralized village-level sourcing model significantly reduces transportation distances, fuel consumption, and associated project emissions.

E. Transportation and Connectivity

Road Connectivity - The site is well connected through:

- Taluka and district roads linking villages to K R Pet town
- State highways connecting Mandya, Mysuru, and Bengaluru

All-weather roads ensure year-round movement of biomass and finished biochar.

Logistics Efficiency

- Average round-trip distance for biomass collection: 30–45 km
- Typical truck payload: 8–10 tonnes
- Reduced congestion compared to peri-urban industrial zones

The location enables optimized logistics planning and low transportation-related emissions, a critical factor in carbon accounting.

F. Site Map

Updated site map will be included

G. Plot Plan

Updated plot plan will be included

FEEDSTOCK RECEIPT, PROCESSING & STORAGE

6. FEEDSTOCK RECEIPT, PROCESSING & STORAGE

Feedstock Receipt

- Tiller trolley capacity of 1T will be used for moving the powdered coconut fronds from the field to the plant storage location
- Trucks capacity of 3T will be used for moving bales of paddy straw & sugarcane trash from the field to the plant storage location
- The vehicle is weighed at the biochar plant's weighbridge.
- Upon arrival at the biochar plant, 2 samples will be collected & the moisture is verified. If the coconut fronds were shredded & dried in the field, they can be fed directly into the continuous pyrolysis reactor without further energy-intensive drying

Processing

- Powdered coconut frond will be pre-dried in the dryer before it is fed into the pyroliser
- Debaling of paddy straw & cane trash, pulverising, grinding & pre-dried in the dryer before it is fed into the pyroliser
- Bamboo slats will be sent to chipping, pulveriser, grinding & pre-dried in the dryer before it is fed into the pyroliser

Storage

- If the moisture content of the biomass is more than the specified range they will be stored in a shed & dried to reduce the moisture content.

Note: for detailed Feedstock Receipt, Processing & Storage process refer to “DPR_Supply Chain Management_KR Pet”

***BIOCHAR
PRODUCTION – MAIN
PLANT & MACHINERY***

7. BIOCHAR PRODUCTION – MAIN PLANT & MACHINERY

The main plant will house the “Continuous Biochar Plant” from a reputed manufacturer technically prequalified by Government of India.

CAPACITY OF THE EACH UNIT:

Each Unit of 2 x 1 TPH lines will produce total 4,800 Tons per annum of Biochar from Agro-forest residues.

FEEDSTOCK (AGRO RESIDUES) REQUIREMENT FOR THE BIOCHAR PLANT AND PRODUCTION OF BIOCHAR:

A) BIOCHAR PLANT:

Estimated Annual capacity of Biochar	: 4,800 Tons / annum / modular unit
Per day Biochar capacity	: 16 Tons / day / modular unit
Based on 24 Hours of operation/day, hourly output	: 2 x 0.4 per line 0.8 Ton/Hr (having <20% moisture) – 2 lines

BRIEF PROCESS DESCRIPTION:

PYROLYSER AND PACKING SECTION

The material from biomass silo shall be taken by screw conveyor to the pyrolyser which will make biochar.

1. The biochar which come out of machine are in hot condition and are then cooled in the cooling section connected to the Chiller unit. Here the hot biochar will pass through the water cooling jackets to cool them to < 50°C
2. The biochar is then fed into the Bagging machine hopper and the packed into 50 kg bags and then stitched and conveyed by slat conveyor to the dispatch section.
3. 1 no. 2.0 TPH bagging machine is planned to cater to the 2 lines of 1 TPH pyrolyzer units

MACHINERIES SELECTED FOR THIS UNIT:

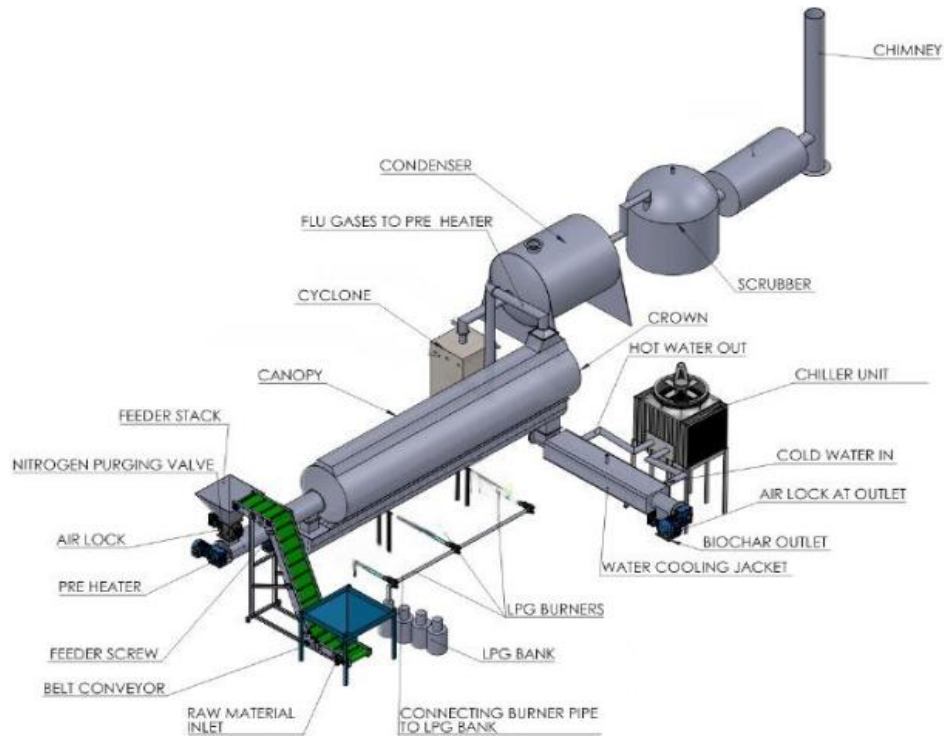
The machineries required for pre-processing, processing is listed below:

- a. Dryer – 1 no
- b. Continuous Biochar Plant - 2 nos (per shed)
- c. Bagging machine – 1 no
- d. Material handling equipment – tipper trolleys (2 nos), wheeled oil drums (2 nos), Drums caddy (2 nos), Drum lifter & tilter (1 no).

Hammer Mill / Grinder (2.0 – 3.0 TPH)	Bagging Machine (2.0 – 3.0 TPH)	Tipper Trolley (1 ton)
		

Drums Caddy	Semi Electric Drum Lifter And Tilter (450 kg)	
		

Main Plant - Continuous Biochar Plant (1 TPH)



MECHANICAL BALANCE OF PLANT (BOP)

8. MECHANICAL BALANCE OF PLANT (BOP)

A. Water systems – sourcing, storage, treatment

The different sources of water are Borewell & Rain Water Harvesting. There is a need to have a Bore well as the plot seems to have a good water table. A suitable point has been identified within the plot. There are bore wells having depth of about 300 - 400 feet with good water source at a distance of about 500 meters from to the plot.

As there will be sufficient roof area over the buildings in the plant rain water harvesting system will be implemented to collect rain water in underground sump.

The sump capacity will be as below:

- Raw water sump – 25,000 litres
- Treated water sump – 25,000 litres
- Fire hydrant sump – 1 lakh litres
- Rain water sump (rooftop) – 1 lakh litres

A **10 KLD Sewage Treatment Plant (STP)** will be setup for water treatment & reuse. Plant design ensures full compliance with Environment requirements & Pollution Control Board norms. Prior consent to Establishment/Permits will be obtained before the plant is commissioned.

1 no 3 HP Pump house is planned for daily use of potable water and **1 no 15 HP Pump house** is planned for Fire hydrant system.

B. Material Handling

Workmen will be provided with uniform, safety gloves, safety helmet, safety glasses, masks, safety shoes to ensure safe & proper material handling.

1 ton Tipper trolley will be used to transport the dry powdered biomass from the storage area to the biochar plant.

1 cu m wheeled oil drums will be used for bio-oil collection & transfer to the settling tanks.

Pneumatic trolleys with pallets will be used to transfer bagged biochar from the bagging machine to the biochar storage area.

C. Fire Protection

A Fire hydrant sump of 1 lakh litres capacity will be commissioned with multiple water source to keep this tank at the maximum level at any given point of time to mitigate any fire risk.

1 no 15 HP Fire hydrant pump will be installed with water outlets located at critical points around the plant for easy access during any emergency. As a standard safety requirement, the fire hydrant system will have all three types of pumps - **Electrical, Jockey, and Diesel**. This setup ensures the system is always pressurized and will work even if the power goes out or a pump fails. Sufficient localized CO2 extinguishers, fire

retardants & firefighting equipment will be provided within the biomass & biochar processing areas.

D. Workshop

Workshop will consist of tools required for maintain all the machineries & material handling equipment's in good working condition.

Optimum level of spare parts inventory will be maintained for immediate attention to any breakdowns to minimise the downtime of machineries

The following template is designed specifically for **bio-feedstock processing and pyrolysis projects**. It ensures that the "Capacity" of the BOP machinery is logically linked to the "Pyrolyser Capacity."

Mechanical BOP Machinery Data Template

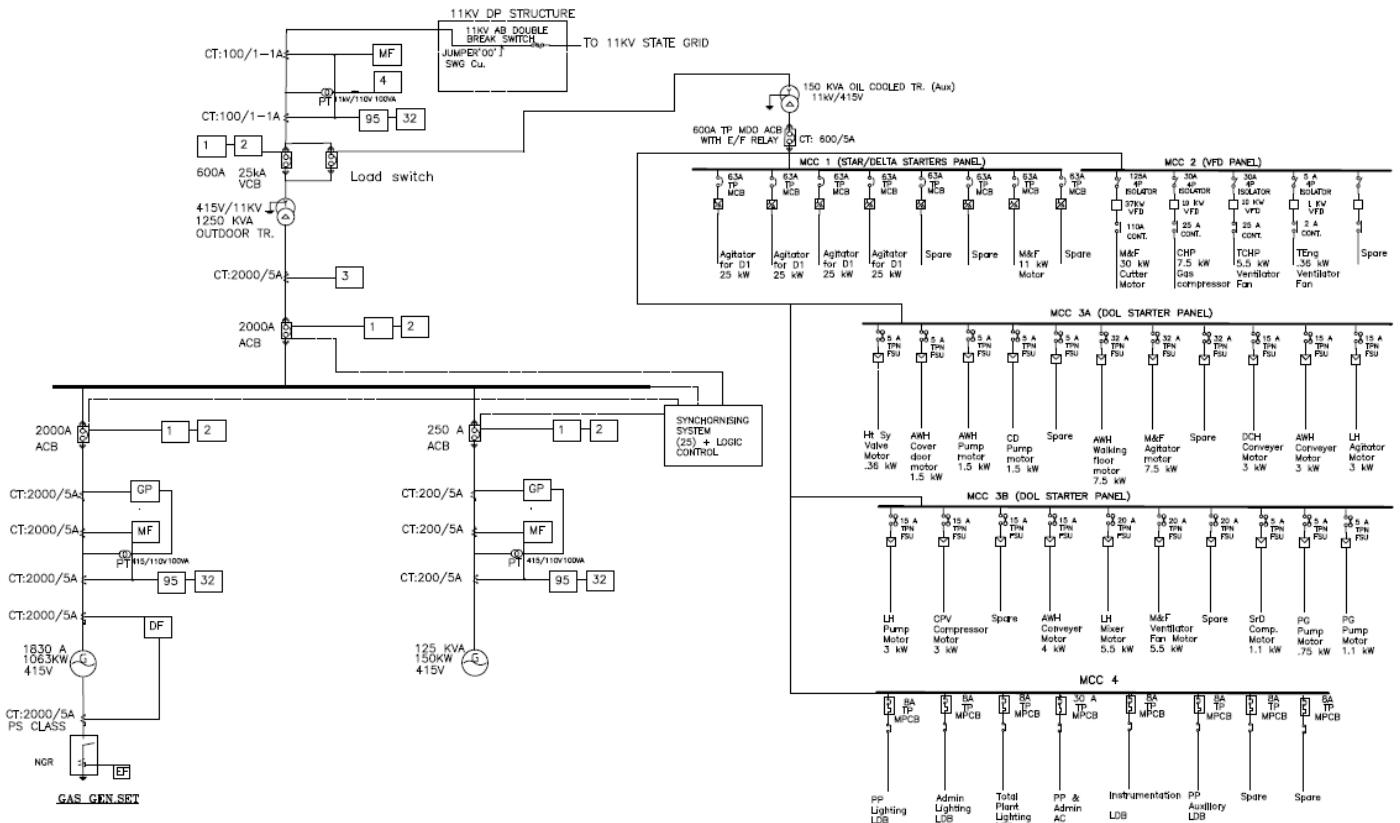
Category	Equipment Name	Manufacturer / Model	Key Specifications / Capacity	Power Requirement	Qty
Material Handling	Electric Tipper Trolley	Loadsmith Industries	1,000 Kg Capacity; 30° Tipping Angle	12 kW Motor (48V)	1
Packaging	Packing & Bag Closing System	Samirika GWP-01	2–3 TPH; Load cell based electronic system	Pneumatic + Electric	1
Core Auxiliaries	Feeder U-Conveyor	Sanjivani Agro	12/14 Feet length	3 HP	1
	Feeder Screw Conveyor	Sanjivani Agro	1.5 Meter length	3 HP	1
	Input Airlock	Sanjivani Agro	Rotary Valve	1 HP	1
	Outlet Airlock	Sanjivani Agro	Rotary Valve	1 HP	1
	Chiller Unit	Sanjivani Agro	50 TR Capacity	-	1
	Water-Cooled Condensers	Sanjivani Agro	1.5 Meter length	-	2
	Cyclone Separator	Sanjivani Agro	500 mm Diameter	-	1
	Discharge Screw Conveyor	Sanjivani Agro	4 Meter (Water Jacketed)	2 HP	1

***POWER SOURCING &
ELECTRICAL
BALANCE OF PLANT
(BOP)***

9. POWER SOURCING & ELECTRICAL BALANCE OF PLANT (BOP)

A. Grid Power Receiving station*

11 KV Receiving station and Transformer - Grid sourcing for loads from 6 pm to 8 am (off peak hours)



*for reference only

B. Captive Power Plant (Solar PV + BESS)

Rooftop Solar PV plant will be installed on all the available roof area of the plant to generate maximum solar RE power.

Plan to have 4 to 6 hours of storage, to be utilized for operating the plant for about 12 hours of solar & stored solar supply during peak hours to minimize the dependency of grid power. Grid power will be utilized during off peak hours to optimize the cost.

Solar PV Rooftop, 3 to 9 KWp, with 2 hours BESS	Solar PV, off grid, 50 to 500 KWp with 2 to 4 hours BESS
	

Solar PV, grid connected, 2.0 to 2.5 MW + BESS	
	

C. Emergency Power

1 no 100 kVA DG Set is planned as standby to supply power and operate the plant for 1 shift during extended power cuts or during disruption in solar power supply.

For emergency loads decentralized UPS is considered at Administration block, Weighbridge & Electrical control room to operate workstations and office peripherals.

D. HT Side (11 kV) & Transformer

- 11kV Outdoor Structure: DP (Double Pole) structure with AB switch and DO fuses.
- Transformer: 100 kVA or 125 kVA, 11kV/415V, Dyn11, Oil Cooled.
- HT Breaker: 11kV Vacuum Circuit Breaker (VCB) or a Load Break Switch (LBS) with HRC fuses (1 No.).

E. LT Switchboards (PCC, MCC & DBs)

The system follows a hierarchical distribution: PCC → MCC → DB.

a) Power Control Centre (PCC) - Main Panel

Component	Rating	Quantity
Main Incomer	160A, 4P, 35kA MCCB (with Microprocessor release)	1 No.
DG Incomer	125A, 4P, 35kA MCCB (Mechanical/Electrical Interlock)	1 No.
Busbars	250A Aluminium/Copper, Tinned	1 Set
Outgoing to MCC	100A, 3P MCCB	1 No.
Outgoing to DBs	32A/63A, 3P MCCBs	2 Nos. + 1 Spare

b) Motor Control Centre (MCC) - For Processing Line

Feeder Type	Rating/Details	Quantity
Main Incomer	125A, 3P MCCB	1 No.
Heavy Motors (Hammer Mill/Grinder)	Star-Delta Starters or VFDs (based on motor kW)	2 Nos. + 1 Spare
Small Motors (Conveyors/Fans)	DOL Starters (Direct-On-Line)	4 Nos. + 2 Spare
Contactors	AC-3 Duty, rated 1.5x motor current	Per Starter
Overload Relays	Thermal or Electronic (Range-specific)	Per Starter

c) Distribution Boards (DB) - Per Building

- Processing Building DB: 63A TPN (Total Phase Neutral) with 6–12 ways for lighting and small power tools.
- Storage/Office DB: 32A TPN with 4–8 ways.
- Spare: 2 spare MCB ways in each DB.

F. Illumination

- Fixture Type: LED High-bay or Well-glass fixtures.
- Ingress Protection: Minimum IP65 (Dust-tight) or IP66 (Water-jet proof for wash-downs).
- Hazardous Area Rating: In the processing/bagging zone, fixtures should be Flame-proof
- Wiring: All cabling must be in sealed GI conduits or armored cables with flame-proof glands.

G. HVAC & Ventilation System

a) Processing Area

- Heavy-duty Exhaust Fans will be mounted at high levels to remove hot air.

- Fresh Air Intakes through wall-mounted louvers with fine mesh filters to prevent insects and debris from entering while allowing cross-ventilation.

b) Dust Extraction

- Suction hoods installed directly over bagging stations and conveyor transfer points.
- Dust Collection by a Bag Filter or Cyclone Separator system to pull carbon dust out of the air before it settles on electrical equipment or is inhaled by staff.

c) Control Room (Climate Control)

- The control room (housing PLC/SCADA) will be maintained at a slight positive pressure to prevent conductive biochar dust from seeping through door gaps and shorting out electronics.
- Standard split AC units with enhanced filtration (HEPA or high-MERV rated) will be installed to handle the ambient dust load.

H. Earthing Protection

- Transformer Neutral: 2 dedicated Copper Plate Earthings.
- Equipment/Body Earth: GI Pipe/Chemical Earth electrodes.
- Grid: All Earth pits should be interconnected via a GI strip (25 x 6 mm or 50 x 6 mm to form an Earth Mat.

I. Lightning Protection

- System: Franklin Rod on the highest point of the reactor building.
- Down Conductors: GI/Copper strips protected by PVC sleeves at the bottom.
- Dedicated Earth Pit: Lightning protection must have its own dedicated earth pit, separate from the equipment earth.

J. Cabling Requirements

- Main Incomer (Transformer to PCC): 3.5 Core, 70–95 sq.mm XLPE Armoured Aluminium Cable.
- Sub-mains (PCC to MCC): 3.5 Core, 50 sq.mm XLPE Armoured Aluminium.
- Motor Leads: Multi-strand Copper Armoured cables (size based on kW, typically 4–10 sq.mm for conveyors).
- Installation: Laid in perforated GI cable trays or underground RCC trenches.

K. Electrical system/control unit

The biochar processing unit consists of the electrical system and the control unit of the biochar plant. There is a control room and a separate MCC room. In the MCC room, there is the switching cabinet. In the control room there is the Automation & control system with display for the biochar plant.

The display is used to select the operating modes (manually or automatically) and to display the operating states, to display data and to generate reports. The system control is fitted with an isolator switch and a system emergency switch-off. Key switches for the drives of the main components are provided so that each drive can be switched off or enabled.

The control room is equipped with air conditioner as well as a fire warning device.



CONTROL & INSTRUMENTATION

10. CONTROL & INSTRUMENTATION

The C&I framework for the modular pyrolysis units is designed to ensure process stability, operational safety, and high-fidelity data logging for Digital MRV (Monitoring, Reporting, and Verification).

A. Distributed Control System (DCS) & Automation

Each modular unit is equipped with a localized PLC-based (Programmable Logic Controller) system with a centralized Human-Machine Interface (HMI). This allows for:

- **Precision Temperature Control:** Continuous monitoring of the pyrolysis chamber via K-type thermocouples to maintain the optimal 500°C – 550°C range. Automated fuel/gas feed adjustments prevent temperature fluctuations that could compromise biochar quality.
- **Residence Time Management:** Automated control of the screw conveyor/rotary kiln speed to ensure a consistent residence time of 45 to 60 minutes, ensuring complete thermochemical conversion.
- **Emergency Shutdown (ESD):** Integrated safety interlocks that automatically trigger system shutdown in the event of pressure spikes or oxygen ingress.

B. Field Instrumentation & Sensing

To provide real-time feedback, the following instrumentation is deployed across the modular unit:

- **Pressure Transmitters:** Monitoring the pyrogas manifold to maintain slight positive pressure, preventing atmospheric oxygen from entering the anaerobic environment.
- **Flow Meters:** Digital mass flow meters to measure the volume of pyrogas condensed into Bio-oil and Wood Vinegar, providing the primary data for yield audits.
- **Level Sensors:** Ultrasonic level sensors in the Bio-oil sedimentation tanks and Wood Vinegar storage drums to manage inventory and prevent overflows.

C. Digital MRV Integration & Data Logging

The C&I system serves as the foundational data source for carbon credit issuance.

- **Cloud Connectivity:** All process parameters (temperature, throughput, and yield) are pushed to a secured cloud server via IoT gateways.
- **Tamper-Proof Records:** Data logging occurs at 1-minute intervals, creating a "Digital Birth Certificate" for every batch of biochar produced to establish the transparency.
- **Remote Diagnostics:** Centralized monitoring allows the engineering team to troubleshoot decentralized modular units in real-time, reducing downtime and maintenance costs.

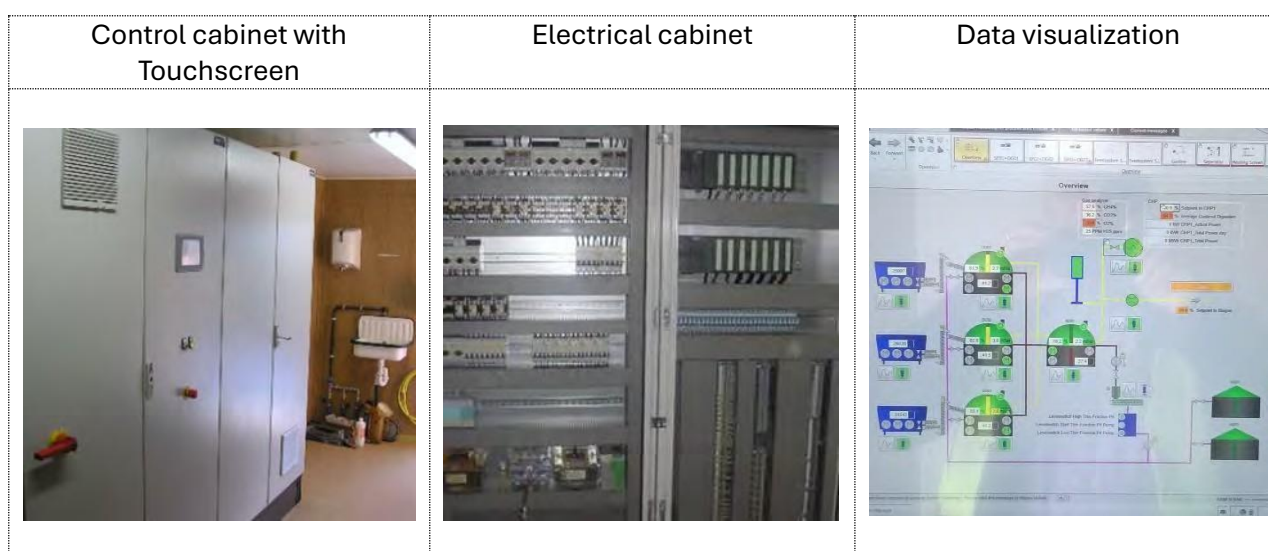
D. Key Parameters for Monitoring (Summary Table)

Instrument	Function	Target Range
Thermocouples	Pyrolysis Zone Temp	500°C – 550°C
Oxygen Analyzers	Process Atmosphere	< 1% O ₂ (Anaerobic)
Variable Frequency Drives (VFD)	Feedstock Flow Rate	Linked to 45–60 min residence
Energy Meters	Auxiliary Power Consumption	To calculate net energy balance

Additional measures to be considered:

- **Calibration Schedule:** all sensors (especially weigh bridges and thermocouples) will be calibrated bi-annually by NABL-accredited labs to maintain data integrity.
- **Power Backup:** UPS (Uninterruptible Power Supply) will be provided to ensure the C&I system remains active during power transitions, preventing data loss or "blind" cooling cycles.
- **SCADA Overview:** Post expansion of 10+ modular biochar plant, a centralized SCADA (Supervisory Control and Data Acquisition) system / control tower will be installed for monitoring

The entire biogas plant will be controlled by a control & automation system supplied by Technology Provider.



CIVIL & STRUCTURAL WORKS, BUILDINGS

11. CIVIL & STRUCTURAL WORKS; BUILDINGS

The civil design for each modular unit focuses on a "Low-Footprint, High-Efficiency" layout. The site is engineered to accommodate the heavy biomass processing machineries & pyrolysis unit, provide weather-proof storage for biomass and biochar, and house the hybrid energy infrastructure.

A. Site Development & Land Utilization

Based on the modular footprint, the site is divided into functional zones to ensure seamless material flow:

- **Leveling & Compaction:** The machinery area will feature a reinforced cement concrete (RCC) plinth (M25 grade) to withstand the static and dynamic loads of the pyrolysis unit, dryer, and bagging machine.
- **Drainage & Pavements:** All internal roads and pathways will be constructed using **Bio-binder integrated asphalt** (thick residue from the process), if available, serving as an in-situ demonstration of the project's sequestration technology.
- **Fencing & Green Belt:** The periphery will be secured with chain-link fencing, with a mandatory green belt consisting of Bheema Bamboo to act as a natural dust and sound barrier.

B. Structural Works & Buildings

To maintain cost-efficiency and modularity, the project utilizes **Pre-Engineered Buildings (PEB)** and industrial-grade sheds:

- **Main Processing Shed:** A high-clearance (min. 6 meters) PEB structure with a Galvalume roof. The shed is designed with open sides for natural ventilation to manage the thermal radiation from the pyrolysis units.
- **Control Room & Lab Office:** A small, insulated prefabricated cabin to house the PLC/DCS control panels, the laboratory equipment for biomass testing, and the digital MRV server.

C. Grid & Renewable Energy Integration (Hybrid Power)

To ensure 24/7 operations while minimizing the carbon footprint of the process, the facility utilizes a hybrid energy system:

- **Grid Connection:** A dedicated transformer and HT (High Tension) / LT (Low Tension) switchgear room to tap into the local grid for base-load requirements.
- **Solar PV Array:** The roof of the processing shed and warehouse will be utilized for a **Rooftop Solar PV system**. This maximizes land use and provides renewable energy for auxiliary loads (chippers, pulverizers, and lighting).
- **Battery Energy Storage System (BESS):** A containerized BESS will be installed to:
 - **Stabilize Power:** Bridge the gap during grid fluctuations or outages, ensuring the C&I system and critical cooling pumps never lose power.

- **Peak Saving:** Store excess solar energy during the day to offset peak-hour grid costs.

D. Earthing & Lightning Protection

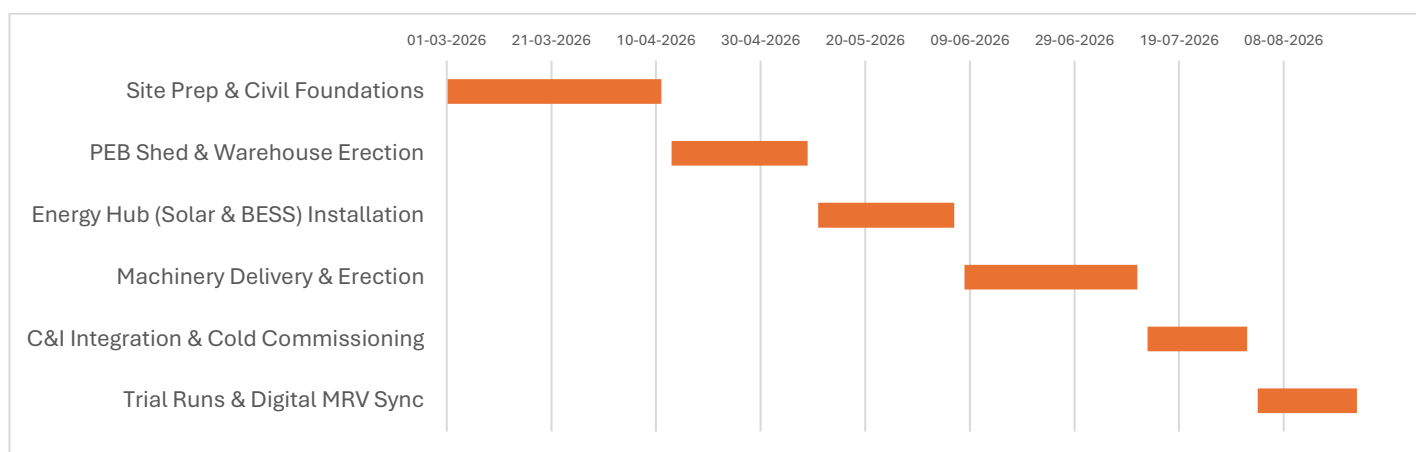
- Given the metallic nature of the equipment and PEB structures, an advanced chemical earthing system and lightning arrestors will be installed to protect sensitive electronic instrumentation.

Summary of Civil & Structural Components

Component	Construction Type	Purpose
Machinery Plinth	Reinforced Concrete (RCC)	Load bearing for >500°C pyrolysis unit
Main Shed	Pre-Engineered Building (PEB)	Weather protection and ventilation
Energy Hub	Prefabricated Container	Housing for BESS and Inverters
Water Tank	Underground / Overhead	Firefighting and Bio-oil cooling loop

Construction Plan for the 1st Modular Biochar Plant (indicative)

Phase	Activity	Start Date	End Date	Duration	Duration	Weeks
Phase 1	Site Prep & Civil Foundations	01-03-2026	11-04-2026	41 days	6 Weeks	01 - 06
Phase 2	PEB Shed & Warehouse Erection	13-04-2026	09-05-2026	26 days	4 Weeks	06 - 09
Phase 3	Energy Hub (Solar & BESS) Installation	11-05-2026	06-06-2026	26 days	4 Weeks	09 - 12
Phase 4	Machinery Delivery & Erection	08-06-2026	11-07-2026	33 days	6 Weeks	13 - 18
Phase 5	C&I Integration & Cold Commissioning	13-07-2026	01-08-2026	19 days	3 Weeks	19 - 21
Phase 6	Trial Runs & Digital MRV Sync	03-08-2026	22-08-2026	19 days	3 Weeks	22 - 24



BIOCHAR PACKING & STORAGE

12. BIOCHAR PACKING & STORAGE

The post-production phase is governed by a rigorous "Track-and-Trace" system to ensure both carbon sequestration integrity and logistical efficiency.

A. Digital Traceability and Standardization

Upon discharge from the biochar unit, biochar is stabilized and packed into standardized **50 kg high-density bags**. Each bag is assigned a **unique, distinctive barcode / QR code** linked to the digital MRV (Monitoring, Reporting, and Verification) platform. This digitization allows for the precise tracking of the carbon footprint from the point of production to the specific GPS coordinates of the farmer's field, ensuring the "permanence" requirements of high-quality carbon credits are met.

B. The "Barter" Model (Paddy and Sugarcane Clusters)

The distribution strategy is to maximise the utilization of the trucks & minimize empty-load transport costs. Biochar bags are transported to sugarcane and paddy clusters via the same vehicles used for biomass collection.

- **The Exchange Ratio:** For every **4 metric tons (MT)** of agricultural residue (paddy straw or cane trash) collected, the project provides **1 ton (20 bags)** of biochar to the farmer **free of charge**.
- **Agronomic Support:** To maximize soil health benefits, the project team provides technical guidance for the co-application of biochar with **Farm Yard Manure (FYM)** at a rate of 4 to 6 MT per acre. This synergistic application enhances microbial activity and optimizes the nutrient-holding capacity of the biochar.

C. Targeted Application in Energy Plantations and Horticulture

For high-value energy plantations and intensive cultivation, the distribution follows a specialized protocol:

- **Sustainable Energy Plantations (Bamboo):** Prior to the planting of Bheema Bamboo, a foundational application of **8 tons (160 bags) per acre** is implemented. This intensive application is designed to rapidly elevate Soil Organic Carbon (SOC) levels. To ensure farmer participation while maintaining project viability, this is provided at a highly subsidized rate of **Rs 200 per 50 kg bag**.
- **Horticulture and Polyhouse Integration:** In polyhouse environments, biochar is utilized as a soil conditioner mixed with vermicompost. This specialized substrate supports intensive cultivation cycles (4 to 6 crops per year), significantly improving moisture retention and crop resilience in controlled-environment agriculture.

BIOCHAR SEQUESTRATION

13. BIOCHAR SEQUESTRATION

Our modular biochar facilities represent a high-impact, circular carbon economy solution designed to convert agricultural waste – specifically coconut fronds, paddy straw and sugarcane trash into permanent carbon sinks while enhancing soil productivity.

Operational Capacity & Output

Each modular plant is designed to process **~14,000 TPA of agricultural residues (as-received basis)** (equivalent to ~8,600 TPA dry basis at 40% average moisture), with a confirmed output of **4,800 TPA** of biochar per year (2×1 TPH lines, confirmed). Sequestration target of **3,333 TPA** of biochar is dedicated to carbon removal agreements. A contingency of 2,200 TPA is allocated for commercial sales (water treatment or metallurgical coke replacement)

Carbon Removal & Credit Generation

The project generates high-integrity carbon credits by leveraging both biochar and bio-oil derivatives. Each modular unit has a potential of **9,360 to 9,750 tCO₂** per year and we aim to contract **6,670 tCO₂ per year** per unit. Calculations are based on a **Permanence Factor of 0.8** and a removal rate of approximately **2 tCO₂ per ton of biochar**. Thick residues from high-temperature bio-oil (6% of feedstock) will be used as a bio-binder in asphalt roads, providing permanent storage similar to biochar and adding **1,920 tCO₂** in sequestration value per plant.

Agricultural Integration & Farmer ROI

Our model incentivizes feedstock collection by returning value to the soil, creating a self-sustaining supply chain. Biochar will be primarily applied to paddy and sugarcane fields at **2 tons per acre**. For sugarcane farmers, a subsidized application cost (approx. ₹8,000/acre) is offset by a **6% yield increase**, recovering costs through the production of an additional 2.4 tons of cane.

We target 3,000 hectares over three years to sequester ~14,400 tons of biochar (4,800 TPA × 3 years). During seasonal shortfall periods (June–August monsoon gap), the primary buffer strategy relies on stored coconut fronds, which are available 365 days per year.

By scaling to three modular plants, the project will achieve **10,000 TPA of biochar sequestration**, removing over **20,000 tCO₂ annually** while providing the carbon offsets necessary for a circular carbon economy.

DIGITAL MRV & DASHBOARD

14. DIGITAL M.R.V. & DASHBOARD

Digital MRV (DMRV) tool support standardized, geotagged data collection across the biochar production value chain.

- a) Feedstock data: Source, type, quantity, moisture level, preparation process
- b) Production data: Production ID, pyrolysis temperature, GPS location, biochar characteristics
- c) Packaging data: QR codes linking each unit to production batch
- d) Application data: GPS location of field application, volume applied, mixed matrix

The core objectives of the proposed solution are:

- Implement full visibility from **biomass collection → biochar production → bagging → field application → SOC improvement → carbon credits**
- Build a scientifically compliant **carbon credit computation engine** (SOC-based, biochar-based, avoided emissions, etc.)
- Provide farmer- and parcel-level traceability to strengthen registry validation
- Ensure tamper-proof, verifiable data trails for carbon audits
- Efficiently manage carbon credit inventory, allocation, and trading
- Enable real-time dashboards and insights for both field operations and management

The solution includes 3 core layers:

A. Operational & Traceability Layer

1) Feedstock Collection & Transport

- Log biomass source details (generator, type, location)
- Capture trip details: loading points, weights, distances
- Estimate transport-based emissions
- Provide source-to-plant traceability

2) Biochar Production, QC & Batch Management

- Create production batches with process parameters
- Record QC attributes (moisture, temperature, residence time)
- Convert output into standardized **50 kg** QR-tagged bags
- Each bag uniquely mapped to:
 - Source feedstock
 - Batch
 - QC data
 - Destination

3) Bag Inventory & Dispatch

- Real-time bag-wise inventory
- Dispatch workflows to farmers / hubs / internal projects
- QR scanning at dispatch & receiving points
- Tamper-proof traceability from origin to application

B. Field Application & Agronomic Layer

1) Farmer & Land Parcel Management

- Maintain farmer profiles
- Record land parcels with GPS coordinates
- Track cropping pattern, soil type, acreage
- Support multiple parcels per farmer

2) Field Application Tracking

- Issue biochar bags to farmers
- Log bag-to-land parcel mapping
- Capture application events:
 - Photos
 - GPS
 - Date/time
 - Quantity per acre
- Mobile/PWA interface for ease of field operations

3) Soil Sampling & SOC Tracking

- Baseline soil tests for each parcel
- Periodic SOC re-tests after application
- Upload lab results & reports
- Compute SOC improvements automatically
- Maintain audit-ready SOC history

C. Carbon Credit & Registry Layer

1) Carbon Credit Computation Engine

The system will generate carbon credits based on:

- SOC delta (pre vs post application)
- Biochar carbon retention
- Avoided emissions (if applicable)
- Adjustment for transport emissions

The methodology and formulae will be configurable based on registry guidelines.

2) Carbon Credit Ledger & Allocation

- Track credits generated per farmer, project, region, season
- Allocate shares to:
 - Farmers
 - Shree Krishan CDR
 - Internal projects or partners
- Maintain a transparent ledger for all carbon credit movements

3) Registry Submission Workflow

- Create registry-ready data packets
- Track validation, verification & certification statuses
- Store and link all supporting documentation required by the registry

- API integration (if registry provides access)

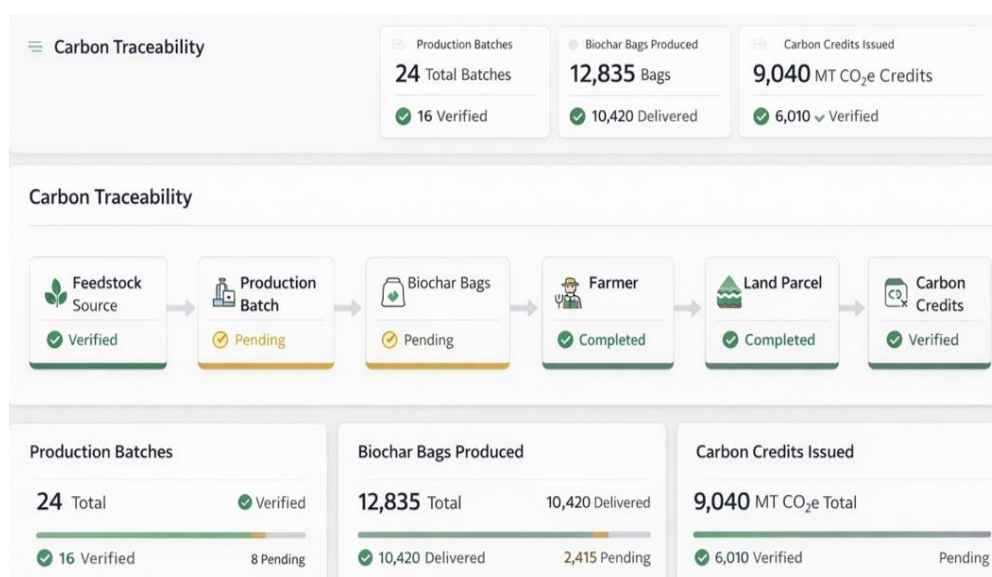
4) Carbon Marketplace / Buyer Management

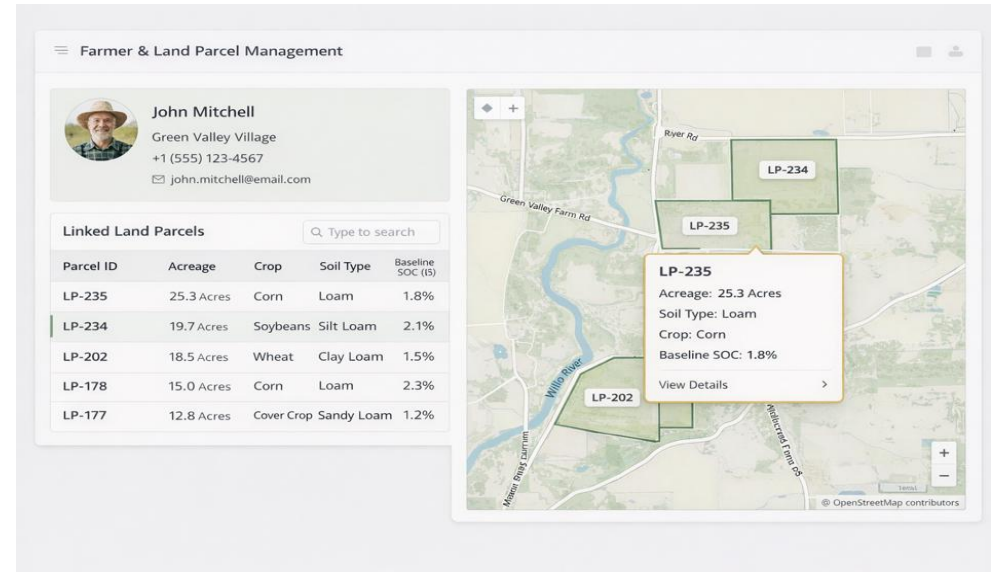
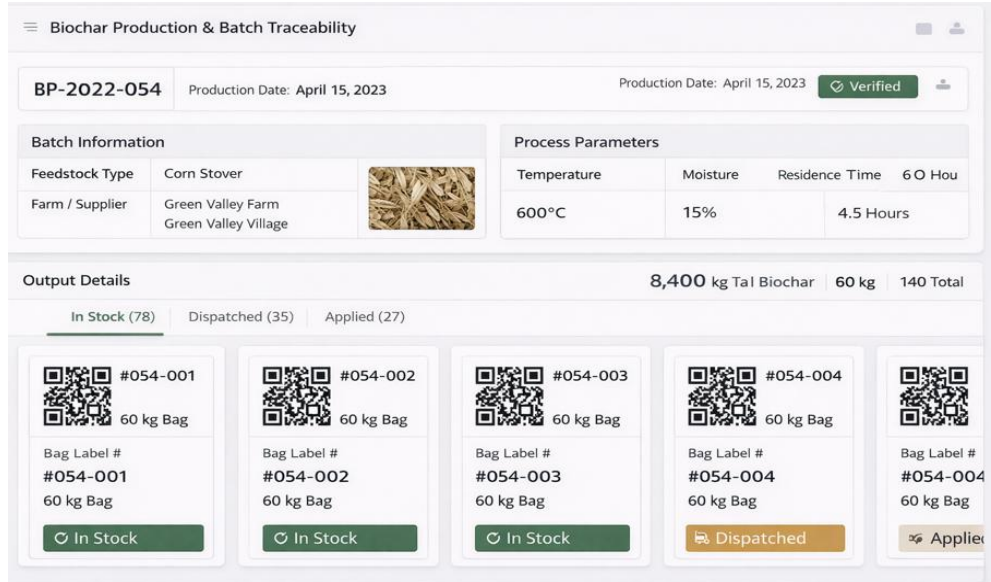
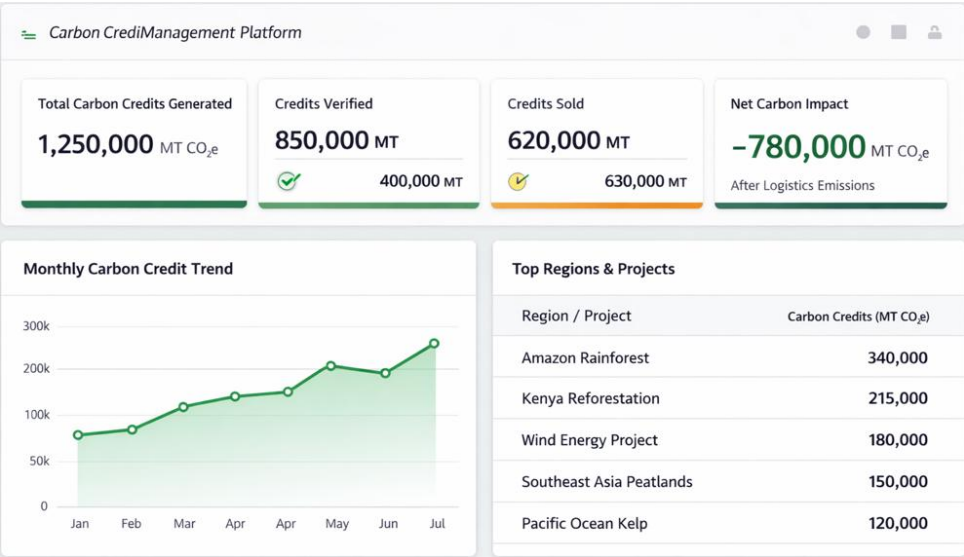
- Maintain inventory of verified credits
- Allocate credits to buyers
- Track transactions and ledger entries
- Generate invoices & buyer documentation

Blockchain-enabled transparency:

- Immutable data from biomass source to biochar sequestration to carbon credits issuance
- Digital MRV and Digitech Technician to collect & aggregate data, validate and time-stamp each step
- REAL-TIME MONITORING PLATFORM – live dashboard for investors & buyers
- Performance metrics: tCO₂e removed, hectares restored, biochar pH & C-content
- Remote auditing, automated reporting for carbon registries

Sample MRV Dashboard screens





HEALTH, SAFETY & ENVIRONMENT MANAGEMENT

15. HEALTH, SAFETY & ENVIRONMENT MANAGEMENT

A. SAFETY POLICY

A. PRE-CONSTRUCTION HSE FRAMEWORK – BIOCHAR PYROLYSIS PLANT (KR PET)

A.1 Statutory Approvals Required (Pre-Construction)

The following statutory clearances must be obtained from Karnataka Pollution Control Board (KSPCB) and relevant authorities prior to commencing civil works: (1) Consent to Establish (CTE) under the Air and Water Acts from KSPCB – for a biomass pyrolysis unit classified under Red/Orange category; (2) Factory License under the Factories Act, 1948 (Karnataka Factories Rules); (3) No Objection Certificate (NOC) from the local Panchayat / Mandya District Administration for land use; (4) Electrical inspectorate approval for HT connection (11 kV BESCOM line); (5) Fire NOC from the Karnataka State Fire & Emergency Services for biomass storage and pyrolysis operations; (6) MSME Udyam registration for the plant entity (Shree Krishan CDR).

A.2 Hazard Identification – Biochar Pyrolysis Operations

The primary hazard categories specific to this plant are: (a) Fire and Thermal Hazards – biomass storage areas, pyrolysis chamber operating at 500–550°C, and hot biochar discharge are the highest fire risk zones. All biomass storage must maintain a minimum 10-metre clearance from the reactor building. (b) Combustible Dust – pulverized coconut fronds and biochar fines create explosion risk. Dust extraction systems (cyclone separators, bag filters) are mandatory at all transfer and bagging points. All electrical fittings in processing areas must be flame-proof (minimum Ex-d rating). (c) Pyrogas Leakage – the anaerobic pyrolysis process generates combustible syngas (CO, CH₄, H₂). The reactor must maintain slight positive pressure, monitored by pressure transmitters with automatic shutdown interlocks triggered at any pressure drop or oxygen ingress above 1%. (d) Heat Stress – operators working near the reactor and dryer will be exposed to radiant heat. Mandatory 30-minute rotation schedules and hydration protocols during peak operations. (e) Ergonomic Hazards – 50 kg bag handling at the bagging station; a semi-electric drum lifter is specified for bio-oil drums. Manual lifting SOPs must be enforced.

A.3 Personal Protective Equipment (PPE) Requirements

All plant personnel must be issued and must wear the following PPE as a minimum: Safety helmet (IS 2925); Safety shoes with steel toe cap (IS 15298); Heat-resistant gloves for reactor and dryer operations; N95 / FFP2 dust masks for biomass processing and bagging zones; Safety goggles (IS 5983) in all processing areas; Hearing protection (earmuffs, SNR ≥25 dB) in chipper and pulveriser areas; High-visibility vest during material handling operations. PPE is provided free of charge by Shree Krishan CDR to all employees and contract workers. A PPE register will be maintained at the site office.

A.4 Emergency Response Plan (Pre-Commissioning)

The Emergency Response Plan (ERP) for the pre-construction and trial-run phases covers:

(1) Fire Emergency – 1-lakh-litre fire hydrant sump with a 15 HP diesel-backed pump ensures water availability regardless of power status. CO2 extinguishers (minimum 5 kg, Type BC) positioned at every 15 metres in processing areas. Fire drill to be conducted monthly. Emergency contact: K R Pet Fire Station (to be confirmed, post-permit).

(2) Pyrogas Leak – Emergency Shutdown (ESD) actuated automatically by pressure/O2 sensors. Manual ESD buttons at the reactor control panel and at the main gate. All non-essential personnel to evacuate upwind.

(3) Medical Emergency – First aid box maintained at the site office. Nearest hospital: K R Pet Government Hospital or private facility]. Ambulance contacts to be displayed at the site gate.

(4) Accident Reporting – All lost-time incidents to be reported within 24 hours to the KSPCB-designated authority and to the Mandya District Labour Office, as required under the Factories Act.

A.5 Environmental Management – Biochar Plant Specific

Air Emissions: The pyrolysis process is designed as a closed, oxygen-free system. Pyrogas is either recirculated as process heat or condensed into bio-oil. Any residual non-condensable gas is combusted in a controlled afterburner before atmospheric discharge. Stack emissions must comply with KSPCB norms for particulate matter (PM ≤ 50 mg/Nm³) and CO (≤ 100 mg/Nm³). Stack monitoring at commissioning and quarterly thereafter.

Wastewater: The plant generates approximately 10 KLD of process and domestic wastewater. A 10 KLD Sewage Treatment Plant (STP) is specified. Treated effluent (to meet KSPCB norms for land application) will be reused for green belt irrigation. Zero liquid discharge (ZLD) is the target.

Solid Waste: Bio-oil sludge and char fines are non-hazardous and will be reprocessed or used as bio-binder. No hazardous waste as defined under the Hazardous & Other Wastes Rules, 2016 is generated by this process.

Green Belt: A mandatory 5-metre-wide green belt of Bheema Bamboo will be established along the entire plant perimeter prior to commissioning, as required by KSPCB and as a natural dust and noise buffer.

1.1 Purpose of Safety Organisation

- This organization is responsible for maintaining safety during the complete contract period of a project. The basic approach to maintain safety is as follows:
- All personnel employed on a project are responsible for their own safety and the safety of personnel next to them.
- Minimization of near-miss accidents can result in minimization of possible actual accidents.

1.2 Responsibilities of Safety Organisation

1.2.1 Site Manager- Safety Manager

Site Manager is responsible for the safety of all activities related to the project. His responsibility is not limited to the safety of the construction work in the field but also for all transportation of personnel, project materials, health and hygienic conditions of work at site and accommodation.

Site Manager is responsible for Safety activities.

1.2.2 Safety Engineer

Safety Engineer is responsible for organizing the safety sites related to the project. He makes timely visits to the construction activities and reports to site manager. Safety engineer receives all data of near-miss accidents from the construction superintendent and analyses those in order to advise the construction superintendent to take necessary action for reducing and minimizing those near-miss accidents.

1.2.3 Construction Superintendent- Safety Superintendent

Construction superintendent is responsible for the safety of the actual activities of all construction for the safety of the actual activities of all construction work, over and above the related activities of work, including all transportation of personnel, project materials, offices and accommodation etc. Construction superintendent gives all necessary instructions to have safe working conditions for personnel and monitors the same during construction period of the project. The construction superintendent implements all requirements of safety and reports and result achieved. All safety documents. Safety manual, safety instructions, safety reports, accidents reports, etc are established by construction superintendent, verified by safety engineer and approved by site manager.

1.2.4 Safety Management Team

Site manager organizes the safety management team which consists of the site manager, safety engineer, and construction superintendent. The safety management conducts safety meetings twice a month to discuss safety requirements, unsafe works found during the past period and to promote safe working and safety requirements for all construction activities until completion of project.

1.2.5 Safety Committee

Site Manager organizes the safety committee which consists of the safety management team plus the quality control engineer, field

planning/progress monitoring engineer, material controller and the medical officer.

Safety committee holds meeting once in a month to discuss safety requirements, unsafe work found on site, suggestion for improving and maintain safe working conditions on construction site. The safety committee makes spot check safety patrol to ensure safety.

1.2.6 Safety Representatives

All persons engaged in any activity of the project are responsible for their own safety and safety of others working in the neighbouring areas. Safety representatives carry out their work within the safety regulations ensuring that they provide no hazards to safety of others and observe the instructions they receive from the safety supervisor.

1.3 Safety Instructions

All personnel engaged on the project are responsible for performing and monitoring all construction and related activities in accordance with the following safety regulations and manuals.

- Fire and safety regulations
 - Minimum health standards in contractor's camp for construction workers
 - Requirement of wearing safety helmets on construction site
 - Safety policy
 - Safety procedures on construction site
 - General safety requirements
 - Work adjacent to existing plant facilities
 - General safety rules
-
- Radioactive material handling and safety requirements.

2.0 SAFETY PROCEDURE

The main purpose of the safety procedure is to carry out work safety in a manner so as to eliminate hazards and risk of death or injury to our personnel, and to third parties on or off the work site and of damage to the works and other facilities and equipment. Also to eliminate the risk to personnel of long term injury due to health hazards. Safety communication chart is attached hereto, are followed for the implementation of Safety procedures.

2.1 Project Manager

Project Manager is responsible for safety of all activities related to the area, not limited to only construction work in the area but also for all

transportation of personnel, project materials, health and hygienic conditions of work at site and accommodation.

2.2 Safety Engineer

Safety Engineer assists in organizing the safety activities. He makes timely visits to the construction sites to check all working conditions of construction activities and any deviation from safety rules and regulation is reported to site - in – charge for correction. Safety Engineer receives all data of accidents from Project Manager and he prepares the detailed safety statistics for submission to Plant Owner. He analyses the safety statistics in order to advise the concerned site-in-charge to take necessary actions for minimizing the frequency of the accidents.

2.3 Site in-Charge

Site-in-charge is responsible for the safety of the actual activities of all construction work and other related activities of work, including all transportation of personnel and materials, and running offices and accommodation camp etc. site- in- charge makes all necessary instructions to have safe working conditions for the work and personnel and monitors these during the entire period of the projects. Site-in-charge receives instructions and advice from the Project Manager and the Safety Engineer for safety and takes necessary measures in accordance with the instructions and reports on the results achieved.

2.4 Safety Supervisors

Every engineer, supervisor, foreman, and other staff including office workers, assumes the duties of safety supervisor for the activities which they are handling or supervising, safety supervisors take all necessary actions for the safety of activities under their supervisions. Safety supervisors in addition to assuming responsibility for their own activities, liaise with other supervisors carrying out work in the same area, to ensure that the activities of one team do not endanger the safety of another team.

2.5 Safety Representatives

All persons engaged in any activity of the project are responsible for their safety of their own actions and for the safety of others working in the neighbouring area. Safety representatives ensure the safety of others and ensure that the instructions received from the safety supervisors are observed.

- 2.6 All work adjacent to the existing plant facilities are carried out strictly under the safety procedure formulated by Plant owner. All necessary

precautions are observed to protect the facility well as to carry our job in a safe way.

- 2.7 Camp and working areas are protected by providing fire fighting arrangements at all possible fire hazard areas. Most of our employees are trained to use fire extinguishers. Periodical training is organized to train our new employees. Regular maintenance of fire extinguishers is carried out.
- 2.8 Periodical fire and safety training Programmes are organized for educating all our employees. Construction hazards and their prevention are highlighted during the training. Some of our employees are deputed to attend the safety training Programmes organized by client or other organizations.
- 2.9 Hard hats, safety shoes, hand gloves, etc., are issued to our employees free of cost for their personal protection. Specific type of safety equipments are provided free of cost to employees as and when required by the employees.
- 2.10 All potential lost time, fatal or vehicle accidents or near miss accidents are immediately reported to Plant owner's safety engineer. Detailed investigation of all accidents is carried out. The causes of accidents and their preventive measures are circulated among all our sites to prevent the recurrences of similar type of accident.

Accident is reported to client through the standard proforma.

- 2.11 Monthly safety statistics are prepared and submitted to the Plant owner through the proforma.

Any specific safety report is submitted as and when required by Plant owner.

Annual summary of safety statistics is submitted to Plant owner.

- 2.12 Any anomalies with regard to safety matters is resolved at site. Specific job safety procedure, if required, is prepared.
- 2.13 Maintenance workshop is established at all our camp sites to carry out periodical maintenance of our vehicles and equipment to keep them in good condition. Periodical vehicle checks are carried out to eliminate traffic accidents. Selected supervisors are deputed to attend the defensive driving course organized by safety organizations or Plant owner to issue our internal company's driving license to successful drivers. A drivers incentive scheme has been introduced.

- 2.14 A good housekeeping committee consisting of the Manager (Project), Safety Engineer, Admin. Officer and medical representative is formed. Housekeeping committee inspects our camp and site facility periodically and inspection report is prepared. Inspection report is sent to the concern site-in-charge for immediately implementation of any suggestions made.
- 2.15 In case of any emergency, immediate actions jointly agreed with Plant owner are taken by the site management has sole authority to prepare emergency action plans.
- 2.16 Any suggestion which helps us to keep our site safe, clean, hygienic and secured is implemented immediately to maintain our good safety standard.
- 2.17 Fire and safety regulations are followed strictly at all our working sites to achieve our goal of accident free completion of project.

Works Adjacent to Existing Plant Facilities

The lighting of fires and the introduction of matches, cigarette lighters and other sources of ignition close to the existing oil and gas facilities is strictly forbidden.

All work, especially hot work, excavation and entry to vessels is controlled by a permit –to-work system.

Industrial Compressed Gas Cylinders

Gas cylinders are transferred, stored and utilized as per the standard safety practices mention in fire and safety rules and regulations.

Excavation

To prevent damage to underground services and to protect the excavating crew, all excavation in or close to company facilities is carried out under the excavation permit procedures formulated by Plant owner.

Work requiring dogging of trenches is programmed so that trenches are not left open for longer than essential.

Personnel working in trenches or pits, excavating soil, clay, sand and similar materials are protected against earth slides, falling objects excessive water build-up trenches. All pits, trenches and excavation are roped off. Warning lights are provided if any excavation is kept open overnight.

Fire Protection and Site Cleanliness

All the works, adjacent facilities, stores, camps and other facilities are protected from damage of fire by providing, inspecting and maintaining suitable types of fire extinguishers.

3.0 GENERAL SAFETY RULES

In addition to the safety regulations mentioned in the previous chapter, the following general safety rules are followed at the construction sites:

- a. Safety helmet and safety shoes are worn.
- b. Supervisor is present at worksite especially where heavy and hazardous jobs are being carried out.
- c. Electrical equipment is considered live unless it is ensured that it is isolated and made dead.
- d. No switch protected with danger board/tag is operated.
- e. Tampering with any types of valves, equipment or moving machinery at site is prohibited.
- f. Water is prevented from falling on electrical equipment
- g. All electrical equipment is properly earthed.
- h. All aisles, passageways, and stairways are kept clean and unobstructed.
- i. It is forbidden to remove guards from moving machinery or to clean/check moving parts of machinery while in motion.
- j. Starting of any machines or any power tools is done only after ensuring that nobody is in danger.
- k. Gas cylinders are kept away from the direct rays of the sun, hot places, and live electric lines. Valves on such cylinder are never lubricated.
- l. It is forbidden to stand under a suspended load

4.0 GENERAL SAFETY REQUIREMENTS

4.1 Introduction and Responsibilities

All work In-charges / Supervisors carry out the work safely to eliminate hazards and to minimize the risk of death or injury to his own personnel, to Client personnel and to third parties on or off the site, and of damage to the works and other facilities and equipments. Supervisors also eliminate the risk to personnel of long term injury due to health hazards.

Supervisor remains fully and exclusively responsible for ensuring that the works are carried out in a safe manner.

4.2 Role of Safety Engineer

To advise on all safety matters, to run safety training, to set up and maintain safety promotion and incentive schemes and to carry out safety inspections. To control and provide liaison on all safety matters with client.

4.3 Safety Training, Inspection and Schemes

Periodical fire and safety training programmes are arranged. Some employees are nominated to attend any specific safety training programmes organized by Plant Owner.

Frequent safety inspections are carried out to find unsafe acts, conditions and concerned supervisors are informed to avoid such unsafe acts and conditions.

Safety Incentive Scheme for the workers has been formulated. Safety Incentives scheme for Drivers has already been formulated and being followed.

4.4 Reporting and Data Gathering

Supervisor immediately reports verbally all potential lost time, fatal or vehicle accidents or near-miss accidents to Safety Engineer for submission to Owner. A detailed investigation report of all lost time accidents is prepared and submitted to the plant owner.

All accidents are reported to Plant owner on a standard Accident Report Form.

Monthly Safety Statistics are submitted to Plant owner on a Standard statistics form.

4.5 Vehicles and Driving

Special attention is given to all matters of road safety with the aim of reducing road accidents and fatalities.

DRIVERS Each driver has a valid Driving License for the type of vehicles he is driving, Maximum speed limit of 20 KMPH is observed and lower speed limits as stipulated in the Plant areas adhered to.

All drivers and front seat passengers wear seatbelts.

Vehicles having adequate capacity for the required duty and which are designed for the service assigned are utilized. No vehicle is driven in an unsafe condition or by an unlicensed driver. All Vehicles are properly identified with the Company's name and serial number. Wheels, tools (including those required to change a wheel), adequate spare fuel, spare water etc., are provided for each journey.

Unless operationally essential, the driver keeps to the good graded or hard topped road system.

- Good housekeeping is maintained. All unnecessary inflammable and other surplus material are removed from the site or facility; Sufficient waste bins are provided on work site.
- Safe storage for all inflammable substances is provided in safe location.
- Parking of vehicles between caravans and portacabins is prohibited.
- All the scrap and rubbish is disposed of at a location and in a manner as approved by plant owner.

4.6 Provision and Operation of Plant, Tools and Equipments

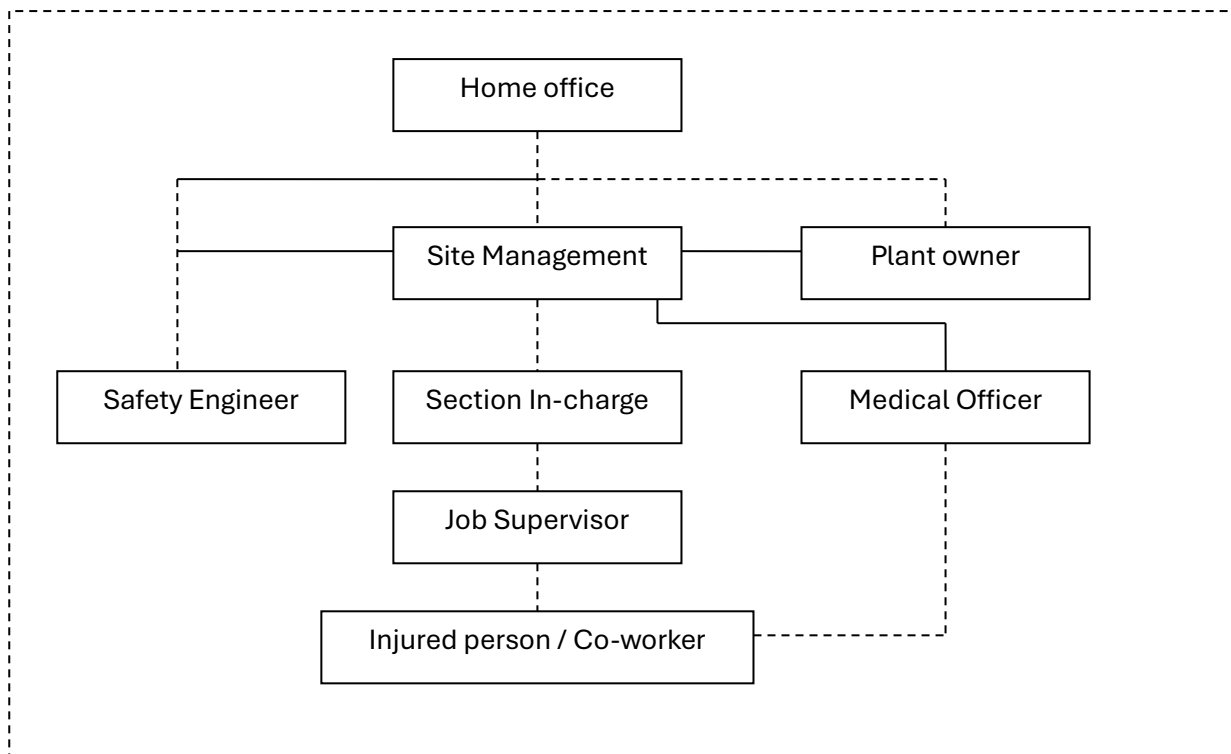
All plant, tools and equipments are maintained in operable condition. Users of the plant, tools and equipment are trained, experienced, and where necessary, licenced or certified. Topmost safety standard is maintained as far as lifting equipments and electrical equipments are concerned.

4.7 Protective Equipments

Protective clothing and equipment are provided to employees and it is ensured that they use these. Adequate eye and body protection is provided to men engaged in welding, burning and grinding operations. Respiration protection equipments are provided to employees working in air contaminated area like shotblasting area, spray painting area etc.

All Safety gear is used as per the demands of the job.

Safety Communication Chart



ACCIDENT REPORT CHART

_____ Reporting Line

----- Communication Line

Environmental Management

5.0 ENVIRONMENT POLICY

5.1 *Policy objectives*

Plant supports clients establish an Environment Policy for their organization which would encompass

- Climate Change Impact Assessment
- Environment Impact Assessment

5.2 *Customer Support Organisation*

Plant provides the above customer support services through its sister company Grameena Abhivrudhi Mandali (GAM) which is a Section 25 Company having mission of sustainable Development of rural communities (refer website: www.gam-rd.co.in).

GAM, as per mandate of Government of India, has developed/is developing various Distributed Generation projects based on locally available agricultural residues/surplus produce, animal waste, food processing/industry waste, etc.

A key element of GAM initiatives are

- Proper understanding of GHG (Green House Gases) mitigation through using Bio Resources / Waste as feedstock for Energy Systems.
- Proper understanding of CHP (Combined Heat & Power) solutions that could be adopted as part of Bio Energy Schemes
- Significant empirical knowledge of producing Organic Fertiliser (using waste from Biomass/Biogas Power Plants) as well as its application for different crops/different soil conditions.
- Significant empirical knowledge in implementing sustainable development projects in rural areas and evaluating socio-economic impact.

5.3 Climate Change Impact Assessment

- Displacement of electricity from the grid by generating electricity from Biomass / biogas and exported to the grid.
- The biomass waste used by the project activity would have been left to decay anaerobically in a solid waste disposal site without methane recovery in the baseline case. Hence, methane emissions are avoided.

5.4 Environment Impact Assessment:

5.4.1 Environmental Assessment Procedure

- Collection of relevant data from various sources such as site visits by a surveyor to the proposed project area, interviews with various stakeholders identified by the project and collection of primary and secondary data from government records on agro-forestry, agriculture and agriculture related industries.
- A desktop review and compilation of this data.
- Environmental aspects (project activities that had potential to impact the environment) of the project will be identified. e.g. Emissions of flue gas from stack, fossil fuel emissions from transportation of biomass, generation of effluent from the water treatment plants etc.

- The environmental aspects will be prioritized based on their potential to have adverse impacts on the project environment and its stakeholders.
- Environmental management plans will be formulated, to control / manage / mitigate these impacts to the best extent possible. The primary aim of the environmental management plan will be to ensure that regulatory standards were met (also reflected in the Power Plant design) and that there exists appropriate monitoring procedures for collection of the relevant data.

5.4.2 Management of Potential Environmental Impacts

- The power plant and auxiliaries shall be ensured to meet the emission norms of the National Ambient Air Quality Standards (NAAQS).
- No radioactive emissions are expected from the proposed Biomass /Biogas plant.
- A green belt, of minimum 5 m width, shall be provided around the periphery of the power plant.

5.4.3 Identification of Socio-economic aspects

Current socio-economic profile of the project area

a. Main income generating activities	
b. Main income generating assets	
c. Average Income of a Farmer from an acre of land per annum	
d. Average level of education	
e. Main types of local transport	
f. Locally available natural resources	

Based on the current socio – economic profile a detailed assessment will be conducted to identify the various potential socio-economic aspects related to the project. These aspects are as listed below:

- Production of quality and reliable power.
- Power generation through a renewable energy technology.
- Activities related to the biomass supply chain i.e. transportation, supply and preparation of biomass.
- Employment requirements at the project site

Socio-economic benefits that will be evaluated

- Increase in productivity of agricultural land and underutilized lands.
- Employment generation through sustainable biomass supply chain
- Grama Panchayats role and involvement.

- SHGs role and involvement.
- Employment generation through organic fertilizer unit and cold storage.
- Employment generation in Biomass/ Biogas Power Plant.
- Employment generation through growth of micro-enterprises, facilitated by availability of assured quality/reliable electric power
- Dissemination of information, to local community, on the advantages of renewable energy based technologies and their applications in creating local value-addition.
- In general, an improvement in the quality of life of the local people, living within the project boundaries.

***ORGANIZATION –
PLANT
CONSTRUCTION AND
O&M***

16. ORGANIZATION – PLANT CONSTRUCTION AND O&M

The organizational structure is designed to support a phased rollout, transitioning from a single pilot module to a multi-unit industrial cluster. The management framework ensures that lessons learned from the commissioning of the first module are instantly integrated into subsequent units.

A. Scaled Commissioning Roadmap

The project follows a rigorous 6-month deployment cycle. A dedicated project team will ensure standardized setup.

Milestone	Module Count	Commissioning Date	Operational Status
Phase I	Module 1	September 2026	Pilot & MRV Baseline
Phase II	Module 2	March 2027	Commercial Scale-up
Phase III	Module 3	September 2027	Cluster Formation
Phase IV	Module 4	March 2028	Regional Expansion
Phase V	Module 5	September 2028	Optimization Phase
Phase VI	Module 6	March 2029	Full Capacity Achievement

B. Operational Philosophy & Shift Structure

To maximize the thermal efficiency of the pyrolysis reactors and meet the 13,000 TPA feedstock target, the plant operates on a continuous cycle:

- **Standard Operation:** 3 Shifts (8 hours each) per day for 6 days a week.
- **Maintenance Protocol:** Sundays are reserved for preventive maintenance.
 - **Day Shift (12 hours):** Deep cleaning of condensers, lubrication of mechanical auxiliaries, and calibration of C&I sensors.
 - **Night Shift:** System pre-heating to ensure the reactor reaches the 500°C threshold by Monday 06:00 AM.

C. Functional O&M Departments

The plant hierarchy is divided into four specialized cells:

1. **Process & Thermal Unit:** Responsible for maintaining the anaerobic environment and residence time. This team monitors the biochar unit core and the bio-oil condensation loop.
2. **Electrical & Mechanical Auxiliaries:** Manages the high-wear components, including the chippers, pulverizers, and screw conveyors.
3. **Control & Instrumentation (C&I):** A specialized team focused on sensor health, PLC uptime, and the integrity of the digital MRV data upload.
4. **Captive Energy Plant Management:** Manages the Solar PV array and BESS. They ensure the battery state-of-health (SoH) is maintained and that the plant switches seamlessly to “Island Mode” during grid outages.

D. Workforce Organization

Each modular unit will be staffed by a blend of technical experts and local “Green Entrepreneurs” incubated at the CoE.

- **Plant Engineer (1 per 2 modules):** Overall P&L and safety responsibility.
- **Shift Supervisors (3 per module):** Ensuring adherence to SOPs and residence time targets.
- **Operators & Helpers (Local Hires):** Sourced from the local community, trained by GAM in feedstock handling and biochar bagging.
- **Maintenance Engineers (2 per cluster):** Specialized in mechanical and C&I troubleshooting.

E. Spares and Inventory Management

Given the 6-month expansion cycle, the project will maintain a **Centralized Spares Hub**.

- **Critical Spares:** High-temperature thermocouples, mechanical seals, and chipper blades are stocked for the entire plant to ensure zero downtime.
- **Standardization:** All modules use identical components (motors, PLCs, valves) to allow for “Plug-and-Play” replacement across any of the six sites.

IMPLEMENTATION SCHEDULE

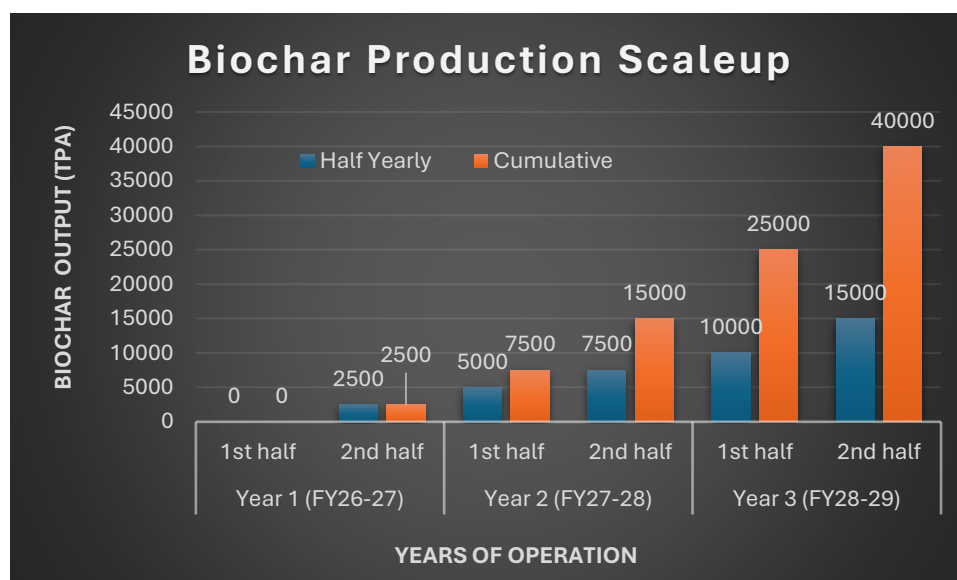
17. IMPLEMENTATION SCHEDULE

A. Biochar Production scale up

Built up over 3 years, during FY 2026-27, 2027-28 & 2028-29:

- End of 6 months: 1 No 1 TPH continuous 2,500 TPA biochar capacity
- End of 12 months: add 1 No 1 TPH continuous 2,500 TPA biochar capacity
- End of 18 months: add 1 No 1 TPH continuous 2,500 TPA biochar capacity
- End of 24 months: add 1 No 1 TPH continuous 2,500 TPA biochar capacity
- End of 30 months: add 1 No 1 TPH continuous 2,500 TPA biochar capacity
- End of 36 months: add 1 No 1 TPH continuous 2,500 TPA biochar capacity

Biochar Output (year-wise) TPA						
Line ID	Year 1 (FY26-27)		Year 2 (FY27-28)		Year 3 (FY28-29)	
	1st half	2nd half	1st half	2nd half	1st half	2nd half
1	0	2500	2500	2500	2500	2500
2	0	0	2500	2500	2500	2500
3	0	0	0	2500	2500	2500
4	0	0	0	0	2500	2500
5	0	0	0	0	0	2500
6	0	0	0	0	0	2500
Half Yearly	0	2500	5000	7500	10000	15000
Cumulative	0	2500	7500	15000	25000	40000



B. Biochar Sequestration scale up

The following table outlines the growth in sequestration capacity, land impact, and carbon credit generation over the next four fiscal years.

Fiscal Year	CO2 Removal Target (tCO2)	Est. Biochar Sequestered (TPA)	Active Modular Units	Land Coverage (Hectares)
FY 2026-27	3,300	~1,650	1 (Pilot/Ramp-up)	825
FY 2027-28	13,300	~6,650	2 Units	3,325
FY 2028-29	26,600	~13,300	4 Units	6,650
FY 2029-30+	40,000	~20,000	6 Units	10,000

Execution Milestones

1. Capacity & Output (The "40k Target")

- **Steady State:** From FY 2029-30 onwards, we will maintain a consistent removal of 40,000 tCO2 per year for a 10-year period (5+5 year contract structure).
- **Biochar Volume:** This requires the sequestration of 20,000 TPA of biochar, derived from approximately 80,000 TPA of agricultural waste (coconut frond, paddy straw and cane trash).
- **Bio-Oil Contribution:** High-temperature pyrolysis will simultaneously yield bio-oil for cash flow, with the "Bio-Binder" residue contributing to the overall permanence factor and CO2 credit volume.

2. Agricultural & Farmer Integration

- **Land Management:** To achieve the 40,000 tCO2 target, we will manage 10,000 hectares of farmland.
- **Paddy & Sugarcane Strategy:** Application remains optimized at 2 tons of biochar per hectare. This cycle ensures a continuous supply of feedstock (1 ton biochar produced per 4 tons of waste collected), creating a closed-loop system with the farmers.
- **Soil Organic Carbon (SOC):** This scale-up will add approximately 15,000 tons of SOC annually across our partner farms, significantly increasing crop resilience and yield.

3. Carbon Credit Portfolio & Energy Blending

- **High-Fidelity Credits:** Based on a Permanence Factor of 0.8, our sequestration methodology ensures high-integrity carbon removal credits that meet international standards (PDD design norms).
- **Energy Synergy:** The 40,000 tCO2 removal capacity allows us to offset the carbon footprint of a significant Variable RE + BESS portfolio, providing firm, carbon-free energy to industrial clients with a high Capacity Utilization Factor (CUF).

Financial Sustainability Note

Each modular unit's feedstock procurement is de-risked by:

1. Bio-oil Sales: 12% yield of feedstock provides immediate liquidity.
2. Carbon Revenue: The contracted CO2 removal (rising from 3,300 to 40,000 tCO2) provides long-term, predictable revenue streams for 10 years.

PROJECT COST BREAKDOWN

18. PROJECT COST BREAKDOWN

The total project cost for the first modular unit (2×1 TPH continuous pyrolysis lines, producing 4,800 TPA of biochar) is estimated at Rs 4.2 Crore (approximately USD 500,000), inclusive of electrical & mechanical BOP, captive solar PV + BESS, civil and structural works, and pre-commissioning. This reflects confirmed vendor quotes from Sanjivani Agro, Ecostan, Samirika Group, and Loadsmith Industries.

The above excludes cost of land and site development.

The feedstock requirement for 2,500 TPA Biochar production, will be 7,500 TPA, with delivered cost of sustainably sourced agricultural residues will around \$ 375,000 @ \$ 50 per ton. Hence, there is economic additionality

Sl. No	Description	Company	Rate / Unit (INR)
1.	Chipper / Hammer Mill	Ecostan	20,23,000
2.	Grinder	Sanjivani	1,01,10,000
3.	Dryer	Sanjivani	
4.	Continuous Biochar Plant	Sanjivani	
5.	Conveyor System	Sanjivani	
6.	Bagging Machine	Samirika Group	4,65,000
7.	Tipper Tractor	Loadsmith	2,64,900
8.	Drum Caddy	Shree Ram Enterprise	27,000
9.	Drum Lifter & Tilter	Shree Ram Enterprise	2,00,00
		Total (INR)	

Note:

1. Applicable taxes extra
2. Transportation charges extra
3. Installation & commissioning charges extra
4. Additional / specialized Training charges extra

PROJECT FINANCIAL OVERVIEW

19. PROJECT FINANCIALS OVERVIEW

Key Financial Assumptions

Confirmed Project Inputs (to be used throughout financial model):

Promoter: Shree Krishan CDR | CAPEX (Module 1): Rs 4.2 Crore (~USD 500,000) | Annual biochar output: 4,800 TPA (confirmed, 2×1 TPH lines) | Annual CDR potential: 9,360–9,750 tCO₂ | Contracted CDR (Year 1 target): ~4,650 tCO₂ (50% of capacity, ramp-up year) | Carbon credit revenue model: Direct B2B offtake agreements with corporate buyers (price per tCO₂ to be confirmed via buyer negotiation; indicative range USD 100–150/tCO₂ per Puro.earth and voluntary market benchmarks as of Q1 2026) | Commissioning: Q4 2026 | Feedstock cost: ~USD 50/tonne delivered (Rs 4,150/tonne); total feedstock OPEX ~Rs 5.8 Cr/year for 14,000 TPA | Bio-oil and wood vinegar revenue: supplementary, not relied upon for base case.

Capacity and Operating Assumptions

1. Installed biochar production capacity: **4,800 TPA (2×1 TPH continuous pyrolysis lines)**
2. Plant availability (steady state): **~90%**
3. Feedstock requirement: **~13,000–14,000 TPA (as received basis)**
4. Carbon removal factor: **~2.0 tCO₂e per tonne of biochar (conservative)**
5. Annual gross carbon removals: **~10,000 tCO₂e**

A. Capital Expenditure (CAPEX) Analysis

CAPEX includes all costs incurred to bring the project to **commercial and carbon-credit-eligible operation**.

Major CAPEX Categories

- Land development and site preparation
- Civil works and buildings
- Biochar reactors and process equipment
- Pre-processing and material handling systems
- Utilities (power, water, fire safety)
- Environmental and pollution control systems
- Digital MRV and IT systems
- Engineering, project management, and contingencies

Indicative CAPEX Breakdown

CAPEX Philosophy: consider procuring equipment's with proven technology, avoidance of experimental or untested equipment and with adequate contingency provisioning

CAPEX Component	% of Total	Remarks
Plant & process equipment	45–50%	Pyrolysis units, controls, Storage, bagging, forklifts
Civil works & buildings	15–18%	Site levelling, drainage, roads, Reactor building, sheds, stores

Utilities & infrastructure	10–12%	Power distribution, water, fire
Pre-processing & handling	8–10%	Shredders, pulverisers, conveyors
Environmental & safety systems	5–6%	Air Pollution Control Devices (APCD), stack, monitoring
Digital MRV & IT	2–3%	Sensors, servers, software
Engineering & contingencies	5–7%	Design, supervision, Cost escalation buffer

CAPEX is phased over Engineering and procurement, Construction and installation & Commissioning and stabilization. *This staged deployment reduces financial risk and aligns cash outflows with project milestones.*

B. Operating Expenditure (OPEX) Analysis

OPEX reflects the **true cost of sustained carbon removal**, including feedstock sourcing, skilled labor, utilities, maintenance, and compliance.

Major OPEX Components

OPEX Component	Description
Feedstock procurement	Biomass collection, transport
Labor & manpower	Operations, maintenance, MRV
Power & utilities	Electricity, water
Maintenance & spares	Preventive & corrective
Consumables	PPE, bags
Compliance & monitoring	Environmental, MRV
Administrative overheads	Insurance, IT, admin

OPEX Characteristics

- Feedstock cost is the **single largest OPEX component**
- Strong focus on **local sourcing** to control transport costs
- Skilled workforce improves efficiency and reduces downtime
- Preventive maintenance minimizes unplanned shutdown costs

C. Revenue Analysis

- Sale of biochar for soil and industrial applications
- Sale of bio-oil for soil, agricultural and horticultural applications
- Sale of tar as binders for road construction applications
- Ancillary services (where applicable)

These revenues are treated **conservatively** and not relied upon for base-case viability.

D. Profitability and Margin Analysis

EBITDA margins are driven primarily by:

- Biochar price realization

- Feedstock cost efficiency
- Plant availability

Cost-to-Revenue Relationship

- Variable costs form the majority of OPEX
- Fixed costs are relatively stable
- Incremental revenue largely flows to the bottom line

E. Cash Flow Analysis

The cash flow model includes:

- Initial CAPEX outflows
- Operating cash inflows from carbon credits
- Ongoing OPEX
- Working capital requirements

Cash Flow Stability

- Carbon credit revenues are periodic (linked to verification cycles)
- Adequate working capital buffers are assumed
- Conservative ramp-up during early years

RISK ASSESSMENT & MITIGATION PLAN

20. RISK ASSESSMENT & MITIGATION PLAN

Risk management is a core requirement for the long-term credibility, bankability, and carbon integrity of the Biochar Project. Unlike conventional industrial projects, biochar-based carbon removal initiatives face **multi-dimensional risks** and managed by classifying the risk.

A. Risk Classification

Risks are categorized into the following groups:

- a) Feedstock and supply chain risks
- b) Technical and operational risks
- c) Environmental, health, and safety (EHS) risks
- d) Regulatory and legal risks
- e) Carbon methodology and MRV risks
- f) Financial and market risks
- g) Social, ESG, and reputational risks
- h) Force majeure and external risks

a) **Feedstock & Supply Chain Risks** cover

- **Feedstock Availability Shortfall:** due to seasonal variability, competing uses, or climatic factors that could reduce biomass availability.

The Mitigation measures considered are multi-feedstock strategy (coconut fronds, cane trash, paddy trash, bagasse, bamboo, mixed residues), village-wise diversified sourcing across villages spread across KR Pet. Stock a buffer inventory of ~30 days

- **Feedstock Price Volatility:** increased demand for residues could drive up procurement costs.

The Mitigation measures considered are long-term sourcing arrangements, localized sourcing to reduce transport cost & continuous productivity improvement in logistics

b) **Technical and operational risks**

- **Technology Underperformance:** Lower-than-expected output or biochar yield could impact production and credit volumes.

The Mitigation measures considered are proven pyrolysis technology, OEM performance guarantees, conservative design capacity assumptions and gradual ramp-up and stabilization period

- **Equipment Failure or Downtime:** Breakdowns could disrupt operations and carbon delivery.

The Mitigation measures considered are Preventive and predictive maintenance systems, critical spare parts inventory, redundancy in key systems and trained maintenance teams

- **Operational Skill Gaps:** Inadequate operator skills could affect safety and consistency.

The Mitigation Measures considered are structured training by **Skills Council for Green Jobs (SCGJ)**, certification and re-certification systems and SOP-driven operations

c) **Environmental, Health, and Safety (EHS) risks**

- **Fire and Thermal Hazards:** Biomass and high-temperature operations pose fire risks.

The Mitigation Measures considered are Fire detection and protection systems, controlled biomass storage, emergency shutdown systems, Regular drills and training

- **Environmental Non-Compliance:** Failure to meet emission or waste norms could lead to penalties or shutdown.

The Mitigation Measures considered are multi-stage pollution control systems, regular environmental monitoring & compliance audits

- **Occupational Health Risks:** Exposure to dust, heat, or noise could affect workers.

The Mitigation Measures considered are Engineering controls, PPE and health surveillance & Job rotation

d) **Regulatory and legal risks**

- **Delay or Change in Regulatory Approvals:** Delays in statutory approvals could impact project timelines.

The Mitigation Measures considered are Early and proactive regulatory engagement & periodic review of compliance requirements

- **Policy or Legal Changes:** Changes in environmental or industrial regulations.

The Mitigation Measures considered are review of Compliance requirements & Continuous policy monitoring

e) **MRV risks**

- **MRV Data Gaps or Errors:** Incomplete or inaccurate data could affect credit issuance.

The Mitigation Measures considered are Digital MRV systems, Trained MRV personnel & Internal audits and cross-checks

- **Permanence and Reversal Risk:** Risk of carbon reversal over time.

The Mitigation Measures considered are Production of stable biochar, Application in appropriate end-uses & Conservative permanence assumptions

f) **Financial and market risks**

- **Cost Overruns:** Unexpected increases in CAPEX or OPEX.

The Mitigation Measures considered are Contingency provisions, Cost control systems & Performance-based contracts

g) **Social, ESG, and reputational risks**

- **Community Opposition:** Potential resistance from local communities.

The Mitigation Measures considered are No land displacement, Local employment generation & Stakeholder engagement mechanisms

- **ESG Non-Alignment:** Failure to meet buyer ESG expectations.

The Mitigation Measures considered are Strong HSE and HR systems, Transparent reporting & Social and environmental safeguards

h) **Force majeure and external risks**

- **Extreme Weather Events:** Floods, droughts, or storms affecting operations.

The Mitigation Measures considered are Robust site design and drainage, Insurance coverage & Emergency preparedness plans

- **Supply Chain Disruptions:** Fuel shortages or transport disruptions.

The Mitigation Measures considered are Local sourcing, Multiple transport options & Inventory buffers

B. Risk monitoring, Review & Governance

a) **Governance Structure**



C. Risk Monitoring & Review Mechanism

- Periodic risk reviews by management (Monthly → Internal, Quarterly → Management review, Annual → Third-party audits)
- Integration with internal audit systems
- Updating mitigation measures as needed

Risk management is treated as a **continuous process**, not a one-time exercise.

REGULATORY & STATUTORY APPROVAL

21. REGULATORY & STATUTORY APPROVALS

The following statutory approvals and registrations are required for the 2×1 TPH Modular Biochar Plant at K R Pet. The project is pre-construction phase as of Q1 2026; all approvals are to be obtained before commencing civil works (Phase 1 target: 01 March 2026).

A. Environmental & Pollution Control

(1) Consent to Establish (CTE) – Karnataka State Pollution Control Board (KSPCB): A biomass pyrolysis unit is classified under the Orange category under the Pollution Control Board's industry schedule. Application to be filed with Mandya District KSPCB office, supported by a site plan, process flow diagram, and stack emission calculations. Estimated processing time: 60–90 days. Status: To be filed.

(2) Consent to Operate (CTO) – KSPCB: To be applied for after plant construction and before commissioning. Supported by stack test reports and STP commissioning certificate. Status: Pending CTE.

(3) Environmental Clearance (EC): Biochar pyrolysis plants with capacity below 5 TPH are generally exempt from Environmental Impact Assessment (EIA) notification under the EIA Notification 2006 (Category B2 threshold). However, confirmation from the State Environmental Impact Assessment Authority (SEIAA), Karnataka, must be obtained in writing before proceeding. Status: Confirmation pending.

B. Industrial & Factory Licensing

(1) Factory License – Karnataka Factories Act, 1948: Required once the plant employs 10 or more workers or uses power. Application to the Mandya District Inspector of Factories, with plant layout, machinery list, and electrical single-line diagram. Status: [To be filed post-CTE].

(2) MSME Udyam Registration: Registration of Shree Krishan CDR as a Micro/Small enterprise under the MSMED Act, 2006. Enables access to priority sector credit, government scheme benefits, and carbon registry eligibility. Status: [To confirm with Shree Krishan CDR]. (6) Trade License – K R Pet Taluka Panchayat: Required for commercial operations within the Panchayat limits. Status: [To be filed].

C. Electrical & Fire Safety

(1) HT Connection Approval – BESCOM (Bangalore Electricity Supply Company): Application for an 11 kV HT connection with a 100–125 kVA transformer. Requires load schedule, single-line diagram, and metering plan. Estimated processing time: 45–60 days. Status: [Application to be filed].

(2) Solar RTS (Rooftop Solar) Registration – BESCOM: Required for grid-connected rooftop solar with net metering. Post-grid connection approval. Status: [Pending BESCOM HT approval].

(3) Fire NOC – Karnataka State Fire and Emergency Services: Mandatory for facilities storing biomass and operating high-temperature processes. Site plan with fire hydrant layout and storage areas required. Status: [To be filed after site plan finalisation].

D. Carbon Registry & MRV Methodology

Carbon Methodology Selection & Project Design Document (PDD): The project must select and register under one of the following CDR methodologies, based on the B2B buyer's registry requirements:

(a) Verra VCS – VM0044 (Biochar Methodology): Suitable for large-scale, multi-buyer arrangements with voluntary carbon market (VCM) listing. Requires third-party validation and verification. Status: To be filed. Target: PDD submission within 3 months of commissioning (by Q1 2027).

ENVIRONMENTAL, ECOLOGICAL & SOCIAL IMPACT

22. ENVIRONMENTAL, ECOLOGICAL & SOCIAL IMPACT

The Biochar Project is designed not only as an industrial facility but as an integrated climate-positive rural development intervention. In addition to delivering high-integrity, long-term carbon removals, the project generates **significant socio-economic, environmental, and ecological co-benefits**.

This section provides a comprehensive assessment of:

- A. Environmental benefits at local, regional, and global scales
- B. Ecological benefits related to soil health, biomass cycles, and land sustainability
- C. Socio-economic benefits to rural communities

A. Environmental Impact

- **Climate Change Mitigation:** The primary environmental benefit is long-term carbon removal through biochar production, Conversion of biomass into stable carbon forms & Permanent sequestration when biochar is applied to soils and Reduction of greenhouse gas emissions from residue burning. This contributes directly to global climate mitigation efforts.
- **Air Quality Improvement:** By diverting agricultural residues from open burning Particulate matter (PM) emissions are reduced, Local air quality improves & Health risks associated with smoke exposure decrease. This is particularly relevant in rural agricultural regions.
- **Waste Management and Circular Economy:** Agricultural waste converted into valuable biochar, process by-products like Bio-oil, Wood vinegar etc reused or repurposed, reduced landfill and open disposal of agri-waste. This model improves resource efficiency.
- **Water Resource Benefits:** Indirect water benefits include improved soil water retention through biochar application, reduced irrigation demand & improved resilience to drought conditions. These benefits enhance long-term agricultural sustainability.

B. Ecological Impact

- **Soil Health Improvement:** Biochar application leads to Improved soil structure and porosity, enhanced nutrient retention & increased microbial activity. These improvements contribute to higher and more stable crop yields over time.
- **Reduction in Soil Degradation:** Reduced erosion due to improved aggregation, improved organic matter stability, enhanced soil resilience under climate stress. This helps maintain long-term land productivity.
- **Biodiversity and Ecosystem Services:** By reducing residue burning and improving soil health, soil biodiversity is enhanced, beneficial microbial populations increase. These ecological gains are gradual but long-lasting.

C. Social Impact

The Biochar project creates substantial **direct and indirect employment opportunities** across the value chain.

- **Direct employment** of 20 – 30 persons, 80% from K R Pet taluka with a mix of skilled, semi-skilled, and unskilled roles.
- **Indirect Employment** of Transporters and logistics providers, Equipment maintenance and service vendors & Ancillary rural services
- **Income Diversification** by monetizing agricultural residues previously considered waste, the project generates additional income streams for rural households, reduces dependence on single-crop incomes & encourages sustainable residue management practices
- **Residue collection** replaces open burning or decay, converting waste into value/energy
- **Skills Development and Human Capital creation** through the **Skills Council for Green Jobs (SCGJ)**, workers receive structured training and certification, Semi-skilled workers transition to skilled roles & long-term employability and income growth. This creates durable human capital assets within the region.
- **Social Inclusion** by Inclusive hiring from rural and marginalized households, Safe, dignified working conditions, no displacement of communities & Gender-inclusive participation where feasible

Data Source:

- a) Dept. of Agriculture Karnataka
- b) Directorate of Economics & Statistics (DES), Karnataka
- c) Sugarcane Development Dept.
- d) Coconut Development Board
- e) District/State Bamboo Mission
- f) Field consultations (FY 2024–25)
- g) <https://mandya.nic.in/en/document-category/district-profile/>
- h) <https://mandya.nic.in/en/district-statistics-at-a-glance-2023-2024/>